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The Effect of Significant IPO Firms on Industry Incumbents: Evidence from Tax Planning

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Abstract. An IPO is a significant event for an individual firm, and the literature examining initial public offering (IPO) firms is extensive. However, there is little evidence on the subsequent effect of a significant IPO on incumbent firms. We extend this literature by using firms' tax policies as a powerful setting to identify whether incumbent firms respond to a significant IPO in their industry. Specifically, we use a first difference analysis to examine whether incumbent firms herd their effective tax rates (ETRs) toward a significant IPO's ETR. We provide robust evidence that incumbent firms adjust their ETR either up or down by one to two percentage points, on average, for up to three years following a significant IPO entrance to the industry. Cross-sectional analyses support attention-based and tax-specific motives for herding, and additional analysis documents market attention-based consequences. Finally, we find that incumbents weigh the benefits of herding against the cost of adjusting their ETR and that incumbent firms use discretionary tax accruals as a less costly mechanism to imitate IPO firms' generally accepted accounting principles (GAAP) ETRs. We contribute to an IPO literature that focuses almost exclusively on the IPO firm. We extend the literature by providing evidence that a significant IPO in an industry can result in a herding response in incumbent firms' policy choices.

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Keywords: IPO • tax benchmark • herding • strategic reactions

1. Introduction

The decision to go public is one of the most important events in a firm's life cycle. Many IPOs generate high levels of media and investor interest and raise substantial capital.¹ Because of the significance of IPOs to U.S. capital markets and the media and public interest in them, a large body of research examines IPO events. However, this literature almost exclusively focuses on the IPO firm itself, including the determinants of its decision to go public, IPO price formation and trading patterns, its long-run performance, and financial reporting decisions around the IPO.² Although they collectively represent overwhelmingly larger market value and market share than the newly public IPO firm, relatively little evidence exists on the effects of significant IPOs on incumbent firms beyond a negative effect on incumbent firms' operating performance following the IPO (Hsu et al. 2010). We extend this literature by examining whether incumbent firms respond to a significant IPO by imitating their corporate policies.³

Because of the "hype" surrounding a significant IPO and its characteristics, investors are often overly optimistic about a significant IPO's prospects and pessimistic about the prospects of incumbent firms (Purnanandam and Swaminathan 2004, Hsu et al. 2010). We posit that incumbent firms will respond by aligning their corporate policies more closely with those of the significant IPO to indicate their quality to stakeholders deciding whether to provide capital to the significant IPO firm or the incumbent firm. Our expectation is grounded in the herding theory of imitative behavior, which indicates that imitation swiftly occurs after the emergence of a focal firm whose actions are weighted more heavily and more positively than others (Lieberman and Asaba 2006, Pollock et al. 2008). Herding can have economic and policy implications because it can multiply the negative effects of undesirable corporate policies or encourage the broad adoption of desirable ones (Lieberman and Asaba 2006). As a result, it is important to understand both the settings in which herding responses occur and why they occur.

Although numerous IPO firm attributes could warrant imitation, we focus specifically on how incumbent firms appear to change their corporate tax policies in response to significant IPO firms' newly disclosed tax policies. Tax policy is a powerful setting in which we can identify incumbent policy responses to significant IPOs for several reasons. First, they are policies for which the outcomes are relevant to, material to, and disclosed by all for-profit firms. Taxes are frequently a firm's largest financial statement expenses and cash outflows. Second, tax policy choices are visible and easily comparable through summary measures (i.e., effective tax rates (ETRs)). Prior research suggests that ETRs allow for easy comparisons between IPO and peer firms and are prominent benchmarks that attract the attention of investors, boards, managers, the media, and watchdog groups (Dyreng et al. 2010, Armstrong et al. 2015, Chyz and Gaertner 2018). Third, research and anecdotal evidence suggest that managers are sensitive to differences between their ETR and the ETRs of their industry peers (Deloitte 2013, Principles for Responsible Investing 2015, Chyz and Gaertner 2018). Finally, corporate tax policy changes that result in either increases or decreases in ETRs can convey credible information because they are not costless.⁴

We examine the ETRs of a sample of 111 significant IPOs and the 12,211 likely affected industry incumbent firm years.⁵ We use a first difference analysis to examine ETR-increasing and ETR-decreasing responses of incumbent firms to the significant IPO. The approach distinguishes this study from those focusing only on ETR reductions (Kubick et al. 2016, Bird et al. 2018, Chen et al. 2019). We provide robust evidence that industry incumbents imitate a significant IPO's cash and GAAP ETRs for up to three years following the IPO year. This imitative behavior involves increasing and decreasing ETRs to align with the significant IPO's tax policies. Our estimations imply an average herding response of between one and two percentage points in either direction, representing a meaningful shift in an incumbent firm's ETR. The moderate economic significance is reasonable and expected, particularly when considering that incumbent firms will increase their ETR to herd.

We perform several cross-sectional analyses to corroborate our primary analysis. Herding behavior occurs after the emergence of a focal firm. Public information releases directing attention to a focal firm's actions exacerbate the bandwagon imitation effect (Bikhchandani et al. 1992, Greenwood et al. 2002). Consistent with information and attention effects of herding behavior, we find that incumbent firms' imitation of a significant IPO's GAAP ETR is more evident when significant IPOs receive more media attention, incumbent firms lose investor attention, and when a significant IPO's prospectus reveals new information. Cross-sectional tests also lend consistent support to tax-specific attention effects,

as the GAAP herding response is more pronounced when an incumbent firm's analysts provide implicit ETR forecasts and during periods of increased corporate tax salience to investors. We also examine whether the herding response varies with tax costs. We find that incumbents are more willing to change their ETR by small amounts than large amounts when herding toward a significant IPO and that, consistent with potential reputational costs from substantial decreases in ETRs, herding is most pronounced for smaller changes in ETR when the incumbent must decrease its ETR to herd and less pronounced when a larger decrease in ETR would be required to herd. We also document a greater upward GAAP and cash ETR herding response for incumbent firms that appear *ex ante* to engage in tax avoidance.

Although our cross-sectional analyses reveal results consistent with the *ex ante* costs and benefits of imitating a significant IPO's tax policies, we also examine the *ex post* benefits to herding. We find that incumbents who herd their GAAP ETR toward the IPO firm's ETR experience a larger increase in investor attention in the month of their subsequent 10-K filing than firms that do not herd. This result is consistent with herding being an effective strategy to regain investor attention following the IPO. We also find that incumbents that herd toward the IPO's GAAP or cash ETR receive additional analyst following as a reward for herding.

Finally, we conduct additional analyses to examine *how* firms imitate the significant IPO's corporate tax policies. Understanding the mechanisms through which incumbents imitate significant IPOs' tax policies should interest regulatory bodies (e.g., the Internal Revenue Service and Financial Accounting Standards Board) whose constrained resources increase incentives to generate ways to identify potential audit targets or financial statement manipulation. Using the cosine similarity of incumbents' disclosures with the significant IPO firms', we find that the tax footnote disclosures of incumbents who herd their GAAP ETR become more similar to the IPO tax footnote. Importantly, we also examine whether the *nontax* footnote disclosures of incumbent firms who herd their GAAP or cash ETR change following the IPO and find that they do not. These findings underscore the importance of aligning tax policies with a significant IPO in an industry. Investments in tax planning, or the unwinding of tax planning structures, often take time to implement and yield changes to a firm's reported tax outcomes (e.g., transfer pricing structures). However, tax accruals are one mechanism for quickly adjusting a GAAP ETR to meet the IPO benchmark. We find that GAAP ETRs, affected by cash tax planning and tax accruals, are incumbent firms' immediate herding responses. Cash ETRs, affected only by cash tax planning, reflect a herding response over a longer time horizon. We also find evidence consistent with using the deferred tax valuation allowance and the reserve for uncertain tax positions to imitate a significant IPO's GAAP ETR.

We perform several tests to rule out alternative explanations and ensure the robustness of our results. We control for pretax performance by including the distance between the IPO and incumbents' pretax cash flows and pretax return on assets (ROA) as control variables and using metropolitan statistical area (MSA), as opposed to industry, to generate significant IPO-incumbent pairings. We continue to find evidence consistent with a tax-specific herding response. We find that incumbents' distance from the industry median ETR is a strong predictor of ETR changes. However, incumbents' imitation of significant IPOs is incremental to incumbents' ETR movements toward industry medians. We also perform additional cross-sectional analysis and rule out industry median reversion as an alternative explanation. We use the accounting quality of the IPO firm to partition our sample and find that the herding response does not vary with it, consistent with a herding explanation as opposed to a learning response by incumbent firms. We confirm that significant IPOs are not necessarily market leaders and results hold when controlling for the distance between the incumbent firms' ETRs and the industry market leader's ETR. Our results are also robust to the effect of expectations of future firm performance on GAAP ETRs. Finally, an inverse probability weighting with regression adjustment (IPWRA) analysis uses a control sample confirming that our results relate to significant IPO events, thus increasing their generalizability.

Our primary contribution is to the IPO literature, which has historically focused on the IPO firm. Hsu et al. (2010) and Spiegel and Tookes (2020) find that industry incumbent firms experience negative returns and a deterioration of operating performance following an IPO in their industry.⁶ We extend this nascent literature by providing evidence of incumbent firms' herding response to policy choices of a significant IPO in an industry. Research has examined herding behavior among individual investors, media, and analysts in the IPO setting (Welch 1992, Rajan and Servaes 1997, Amihud et al. 2003, Pollock et al. 2008); however, to our knowledge, no study has examined herding behavior among firms following an IPO.

In related work, Chen et al. (2019) find that incumbent firms decrease ETRs after an industry IPO, suggesting a "race to the bottom" for ETRs for firms to maintain market share. In contrast, we appeal to the herding theory that suggests a broader imitation response by incumbent firms, resulting in increases or decreases in ETRs to move closer to significant IPOs' ETRs. The "leader-follower" imitation effect to a significant IPO that we document differentiates our findings from Armstrong et al. (2019), who do not examine IPOs and are only able to document a "general diffusion" of tax strategies among peers. Our results complement and extend Armstrong et al. (2019) because the authors find no evidence that industry peers

follow a focal firm, although they directly test it. Our results suggest that a significant IPO firm becomes a focal firm shaping industry norms for tax policies. Also, Armstrong et al. (2019) focus only on infrequent tax law changes, and the IPO setting allows a more frequent identification opportunity to observe an influential peer's ETR. Finally, studies often find evidence that firms use discretionary changes in the deferred tax valuation allowance and reserve for uncertain tax positions to meet analyst forecasts. We find that incumbents use these discretionary changes to imitate the tax outcomes of their peers.

2. Hypothesis Development

The IPO literature is extensive, and three primary research questions categorize it. The first question asks why a firm decides to go public. The second asks how investors price an IPO's initial issuance, and the third examines the consequences to the IPO firm of going public. One unique aspect of the IPO literature is its almost exclusive focus on the IPO event's causes and consequences for only the IPO firm. The IPO literature's focus on the IPO firm is somewhat myopic, given that an IPO represents a shock to the many existing public firms in the shared industry. However, the IPO literature reveals surprisingly little about a significant IPO's effects on the industry's incumbent firms.

Hsu et al. (2010) examine the consequences of a significant IPO to industry incumbents. They find that completing a significant IPO has an immediate, negative effect on existing firms' stock prices in the industry. Industry incumbents also experienced declining operating performance in the three years following the significant IPO.⁷ Hsu et al. (2010) explain that understanding the effects of a significant IPO can help incumbent firms "understand how the new issuance affects their competitive environment and how they can strategically respond to it" (p. 496). Spiegel and Tookes (2020) extend Hsu et al. (2010) by focusing on the reasons underlying the decline in industry incumbents' operating performance following an IPO in their industry and attribute this pattern to downward industry trends instead of the IPO's competitive effect in the industry. We extend both Hsu et al. (2010) and Spiegel and Tookes (2020) by focusing on whether and how incumbent firms strategically respond to a significant IPO.

A significant IPO within an industry introduces high levels of environmental uncertainty as investors, incumbents, and other market participants cannot reliably predict the effects of the IPO on their prospects. When uncertainty is high, herding theory suggests that it can be rational for individuals or organizations to ignore their private information and preferences and imitate others' actions (Pollock et al. 2008).⁸ Under this herding theory of imitation, news and recent information about a single firm can dominate the majority's attention and influence others' beliefs and actions. Firms seeking to

reduce uncertainty and gain approval will imitate a focal firm's most readily available information and socially acceptable attributes (Pollock et al. 2008). Several studies acknowledge the existence of herding behavior among individual investors in the IPO setting. Both Welch (1992) and Amihud et al. (2003) find that investors, unsure about valuing the previously private IPO firm, follow other investors' actions when deciding whether to purchase IPO shares. Furthermore, studies also provide evidence consistent with herding behavior in media coverage (Pollock et al. 2008) and analysts following an IPO (Rajan and Servaes 1997). No study, to our knowledge, has examined herding behavior among firms following an IPO.

An IPO firm and its investment banker have incentives in the pre-IPO period to attract as much attention to the IPO as possible (Cook et al. 2006). Media coverage legitimizes firms' attributes by altering stakeholder perceptions of their appropriateness and desirability. Pollock et al. (2008) find that the public attention focused on the IPO firm influences investors' acceptance of the IPO firm's characteristics and decreases risk perceptions, thereby reducing the IPO's underpricing. Research also suggests media attention increases IPO volume, return volatility, and postoffer performance (DuCharme et al. 2001, Demers and Lewellen 2003, Liu et al. 2014a,b, Bushee et al. 2020).

We expect the attention focused on the significant IPO by industry stakeholders establishes it as a high-status focal firm receiving support from media legitimization of its characteristics and investors who often rush to purchase the IPO's shares. Stakeholder capital flows more freely to the significant IPO than to the industry incumbents, as investors overestimate the new IPO's prospects and underestimate industry incumbents' prospects (Hsu et al. 2010). This alignment of capital flow with the new IPO could lead to incumbent firms' strategic herding response. Although the IPO firm was a competitor to incumbent firms before its IPO, the incumbent firms and their investors lacked publicly available information against which to compare the IPO firm's policy choices. The issuance of the IPO prospectus is when the significant IPO is the most visible to industry incumbents. It is also when the most information about the significant IPO becomes public in a primarily standardized form (Dambra et al. 2019). Thus, a significant IPO represents a potential shock to industry norms and a source of visible and desirable firm characteristics.

Incumbent firms are likely to imitate several significant IPO attributes. We focus on corporate tax policy because it represents a powerful setting in which we are more likely to identify incumbents' responses. In contrast to many corporate policies where firms decide to adopt or not adopt the policy, all corporations are subject to the corporate income tax. Tax planning is a primary consideration in the IPO process because it represents a high-risk area for financial reporting and tax. As a result, practitioners release vast guidance about the importance of tax

planning leading up to the IPO, substantiating an IPO's ETR as a legitimate benchmark for incumbent firms. For example, E&Y's IPO guide stresses the importance of tax disclosures in the IPO prospectus, stating, "[e]ffective tax structure and reporting are key, especially with the rise in demand for transparency worldwide."⁹

Furthermore, the herding behavior we expect from incumbents primarily arises from the dominance of information that is easy to recall and easy to use (Pollock et al. 2008). The ETR in an IPO's prospectus is an obvious and easily understood summary measure of the cumulative effect of the IPO's tax policies. Tax policies are also a meaningful comparison point across firms because they are economically significant to their cash flows and operating performance. Tax policies also affect other important firm attributes salient to investment decisions, such as the cost of debt and equity and firm value (Desai and Dharmapla 2009, Hasan et al. 2014, Goh et al. 2016, Cook et al. 2017). Investors likely view corporate tax policies that are less aggressive than industry peers negatively because the firm has fewer cash flows to invest in positive net present value (NPV) projects than other firms in the industry.

Although tax savings are an important benefit, strategies that reduce the ETR are also often accompanied by costs associated with a lack of business purpose, reputation risk, detection risk, and the risk of adverse attention effects (Graham et al. 2014). Evidence suggests that management is sensitive to these risks. A 2014 survey by EY reports that eighty-nine percent of the largest companies they survey are "somewhat or significantly" concerned about media coverage of their firm's taxes.¹⁰ The survey of Graham et al. (2014) indicates that 69% of corporate tax executives list reputation as one reason firms may not adopt tax avoidance strategies. It also notes that many executives cited a "Wall Street Journal test" in evaluating tax planning strategies, abandoning them if they would "look bad" on the newspaper's front page. Activist groups, such as Institute on Taxation and Economic Policy (ITEP) or Citizens for Tax Justice, often highlight corporations that pay little or no tax. These publications tend to elicit reactions from lawmakers to curb perceived tax avoidance activities. A recent law review article and anecdotal evidence indicate that ESG issues, including the appearance of tax avoidance, are important to investors in the IPO process (Labbe 2016). A significant IPO can establish a visible benchmark that stakeholders, the media, and lawmakers use to compare incumbent firms, potentially exacerbating the costs of being below the new benchmark and being seen as a tax-avoiding company.

Although empirical evidence documenting reputational costs of tax avoidance is mixed (see Asay et al. 2021 for a discussion of this literature), academic studies document other costs associated with tax avoidance. For example, Kubick et al. (2016) find that tax-avoiding firms are more likely to receive a tax-related Securities and Exchange Commission (SEC) comment letter and that

these firms subsequently decrease their tax avoidance. Stakeholders are also likely to negatively view firms with more aggressive tax policies than industry peers because tax avoidance creates uncertainty about future cash flows associated with potential IRS audits (Henry 2018). This greater uncertainty can result in increased stock price crash risk (Kim et al. 2011), a higher cost of debt financing (Hasan et al. 2014), and higher fees paid to the firm's external auditors (Donohoe and Knechel 2014). Hanlon and Slemrod (2009) find that revelation of involvement in tax shelters, a particularly aggressive tax avoidance strategy, is associated with stock price declines. Importantly, several studies also suggest that firms suffer adverse consequences of being different from industry tax avoidance norms in *either direction*, such as a higher ex ante cost of capital or forced chief executive officer (CEO) turnover (Cook et al. 2017; Chyz and Gaertner 2018).

To summarize, we hypothesize that a significant IPO becomes a focal point for the media, investors, and incumbent firms in an industry in the pre-IPO process that legitimizes its corporate tax policy post-IPO. As a result, incumbent firms alter their corporate tax policies to align with the reference point set by the significant IPO to reduce uncertainty arising from the significant IPO's effect on capital flow and the prospects of their industry. However, we acknowledge that IPO prospectus disclosures may be of lower quality due to the incentives of the IPO firm to present favorable information (Lo 2008). It is, therefore, possible that incumbents will not herd toward IPOs' disclosures. This discussion leads to our hypothesis, which we state in the alternative form.

Hypothesis 1. *Incumbent firms corporate tax outcomes herd toward those disclosed by significant IPO firms.*

Our expectation is partly motivated by Armstrong et al. (2019); however, there are some important differences in the theory we use to motivate our hypothesis and the settings in which it applies. Armstrong et al. (2019) use game theory when developing expectations for whether firms will alter their tax policies in response to changes in peer firm ETRs arising from tax law changes. When Armstrong et al. (2019) test whether industry peers respond to a focal firm, they find no evidence of a "follow the leader" dynamic following tax law changes. In contrast to tax law changes that happen only rarely, we examine whether the more common occurrence of a significant IPO in an industry can also elicit an imitation effect. Our expectations are also rooted in an alternative theory of herding behavior that relies on the existence of a focal or leader firm and attention directed toward it. Thus, while other tax studies provide evidence of an imitation effect, we view them as motivation for our focus on tax policies in the IPO setting but do not necessarily expect their conclusions to extend to the IPO setting.

3. Data, Measures, and Research Design

3.1. Sample

To test our hypothesis, we examine whether incumbent firms adjust their tax policies in response to observed differences between their own and those of a significant IPO. Like Hsu et al. (2010), we limit our identification of IPOs in each industry to "significant IPOs" to isolate only those we expect to elicit a herding response from incumbent firms. We obtain a list of domestic IPOs by nonfinancial registrants from 1993 to 2018 from Thomson Financials' Securities Data Company (SDC) database. We start the sample period in 1994 because our research design requires that we obtain lagged tax expense data computed under Accounting Standards Codification 740, enacted in 1992 (Kubick et al. 2015). After dropping the IPOs with missing Compustat or the Center for Research in Security Prices (CRSP) data, 5,138 IPOs in 65 distinct industries remain. We identify significant IPOs by choosing IPOs within a two-digit Standard Industrial Classification (SIC) code with the highest gross proceeds three years before and three years after the IPO date (Hsu et al. 2010).¹¹ This selection criterion addresses the concern that other concurrent IPOs from the same industry may influence the effect of the significant IPO on incumbents.

Although the number of significant IPOs is small, the number of incumbent firms they potentially affect in an industry is large.¹² Furthermore, significant IPOs are also more likely to be certified by top investment banks and audited by Big N audit firms, increasing the perceived legitimacy of their policies. Consistent with prior literature, we exclude IPO firm-years with negative total tax expenses, cash taxes paid, or negative pretax book income before special items (Dyreng et al. 2008, Kubick et al. 2015). We focus only on profitable significant IPOs to ensure that their ETRs, our corporate policy measures of interest, are interpretable.¹³ This focus leaves us with a sample of 111 significant IPOs from 1993 to 2018.¹⁴

Using two-digit SIC codes, we then generate a group of industry-matched incumbent firms for each significant IPO.¹⁵ We drop firms incorporated outside the United States and exclude incumbent firm-years within three years of its IPO event because newly public firms possess competitive advantages and respond differently to significant IPO events than seasoned public firms. We keep firm-years in a three-year window following a significant IPO in their industry for the remaining incumbent firms. We exclude incumbent firm years with negative total tax expenses, cash taxes paid, or pretax book income before special items consistent with our significant IPO selection criteria. Last, we require all observations to have nonmissing values for regression covariates. Our final sample, detailed in Table 1, consists of 12,211 firm years matching significant IPOs in 53 two-digit SIC industries. Panel A of Table 1 provides the sample selection details, whereas Panel B lists the frequency of observations by year.

Table 1. Sample Selection and Distribution

Sample selection process			Panel A: Sample selection			# of firm-years
Compustat domestic nonfinancial service firm-years from 1994 to 2018						138,983
Drop:						
Three years before or three years after the firm's IPO						(21,849)
Not within three years after a major IPO from a peer company in the same industry						(68,096)
Incumbent and IPO firms with negative total tax expense, cash taxes paid, or negative pretax book income before special items						(31,723)
Missing values for the regression variables						(4,903)
Incumbents associated with IPOs taxed as pass-through entities						(201)
Final sample						12,211
Panel B: Sample distribution						
Significant IPOs			Incumbent firms			
IPO year	Frequency	Percent	Fiscal year	Frequency	Percent	
1993	1	0.90				
1994	4	3.60	1994	119	0.97	
1995	4	3.60	1995	360	2.95	
1996	7	6.31	1996	885	7.25	
1997	6	5.41	1997	836	6.85	
1998	3	2.70	1998	558	4.57	
1999	5	4.50	1999	260	2.13	
2000	1	0.90	2000	262	2.15	
2001	7	6.31	2001	475	3.89	
2002	5	4.50	2002	538	4.41	
2003	2	1.80	2003	281	2.30	
2004	9	8.11	2004	731	5.99	
2005	6	5.41	2005	950	7.78	
2006	9	8.11	2006	1,079	8.84	
2007	5	4.50	2007	540	4.42	
2008	1	0.90	2008	373	3.05	
2009	5	4.50	2009	409	3.35	
2010	1	0.90	2010	497	4.07	
2011	3	2.70	2011	393	3.22	
2012	3	2.70	2012	544	4.45	
2013	8	7.21	2013	529	4.33	
2014	3	2.70	2014	543	4.45	
2015	3	2.70	2015	473	3.87	
2016	4	3.60	2016	251	2.06	
2017	3	2.70	2017	137	1.12	
2018	3	2.70	2018	188	1.54	
Total	111	100.00	Total	12,211	100.00	

Notes. This table presents the sample selection process in Panel A and the frequency distribution of firms across our sample period. Our final sample is 12,211 firm years matched with IPOs in 53 two-digit SIC industries. Our sample includes 111 significant IPO events during 1993–2018 and incumbent firms with requisite data matched on two-digit SIC codes. The online appendix provides the 111 significant IPOs, their IPO year, and two-digit SIC. Panel B presents the IPO frequency distribution and matched incumbent firms by year.

3.2. Research Design and Key Variables

Our first measure of corporate tax policy is *CASHETR*, equal to cash taxes paid during the year divided by pretax book income minus special items. Our second measure is *GAAPETR*, equal to total tax expense for the year divided by pretax book income minus special items. The former reflects tax planning strategies that affect a firm's cash taxes. In contrast, the latter reflects both tax planning strategies that affect cash tax payments and the effects of tax-related accounting accruals (e.g., the deferred tax valuation allowance and the reserve for uncertain tax positions). Consistent with prior literature (Dyreng et al. 2008, Cen et al. 2017), we winsorize *CASHETR* and *GAAPETR* at zero

and one. We focus on ETR measures as proxies for corporate tax policies rather than book-tax differences (Henry and Sansing 2018) because they are the most visible and salient measures to external parties (Chyz and Gaertner 2018).

Our ordinary least squares regression models the association between the change in an incumbent's ETR and the distance between an incumbent ETR and the significant IPO's ETR in the prior year as follows:

$$\begin{aligned} \Delta ETR_{i,t} = & \beta_0 + \beta_1 \text{Diff_LagETR_IPO}_{i,t-1} \\ & + \beta_2 \text{Diff_LagETR_Median}_{i,t-1} \\ & + \theta \Delta \text{FirmControls}_{i,t} + \delta_t + \varepsilon_i. \end{aligned} \quad (1)$$

Our dependent variable, $\Delta ETR_{i,t}$, is incumbent firm i 's one-year change in the cash or GAAP ETR from year $t - 1$ to year t . We define $\Delta ETR_{i,t}$ for an incumbent firm in each of the three years following the IPO event. Our independent variable of interest, $Diff_LagETR_IPO_{i,t-1}$, is an incumbent firm i 's cash ETR (GAAP ETR) from year $t - 1$ minus the significant IPO's cash ETR (GAAP ETR) from year $t - 1$. Because the effects of newly implemented tax policies can take time to yield changes in ETRs, our research design allows for nonstatic IPO ETRs in the years following the significant IPO and a reasonable time for incumbents to adjust their tax policies to imitate the significant IPO's tax policies. Specifically, our dependent variable and our independent variable of interest are both rolling distance measures over three subsequent periods. A negative $Diff_LagETR_IPO_{i,t-1}$ coefficient would indicate an incumbent firm with an ETR higher (lower) than the significant IPO's ETR in year $t - 1$ reduces the distance between its ETR and the significant IPO's ETR in the following period by adjusting it down (up).

We include a vector of industry and firm controls based on prior literature (Kubick et al. 2015, Cen et al. 2017, Bird et al. 2018). Importantly, we include the rolling distance between an incumbent firm's ETR and the industry median ETR in year $t - 1$ ($Diff_LagETR_Median_{i,t-1}$) to control for the possibility that incumbent firm ETR changes represent a reversion to broader industry benchmarks.¹⁶ We use the industry median instead of the mean because outliers affect it less. We also include incumbent pretax operating performance ($\Delta ROA_{i,t}$) because Hsu et al. (2010) find that a significant IPO can affect incumbent performance, and ETRs correlate with performance.¹⁷ Additional control variables include changes in research and development expense ($\Delta RD_{i,t}$), advertising expense ($\Delta AD_{i,t}$), selling, general, and administrative expense ($\Delta SGA_{i,t}$), capital expenditures ($\Delta CAPEXP_{i,t}$), cash and cash equivalents ($\Delta CASH_{i,t}$), foreign income ($\Delta FI_{i,t}$), equity method earnings ($\Delta EQINC_{i,t}$), presence of net operating loss ($\Delta NOL_{i,t}$), intangibles ($\Delta INTAN_{i,t}$), property, plant, and equipment ($\Delta PPE_{i,t}$), free cash flow ($\Delta FCF_{i,t}$), size ($\Delta SIZE_{i,t}$), and leverage ($\Delta LEV_{i,t}$). The Appendix provides detailed variable definitions. We control for year fixed effects (δ_t) and cluster standard errors by firm when estimating Equation (1).

3.3. Descriptive Statistics

Panel A of Table 2 presents descriptive statistics for our sample. We use differences/changes in our variables for our analyses and tabulate the descriptive statistics for their levels. On average, the distance between incumbents' and significant IPOs' GAAP ETRs is close to that between their cash ETRs. Incumbent firms have ETRs lower than significant IPOs' ETRs on average. Because our sample has incumbent firms with ETRs above and

below significant IPO tax rates, the means of our distance metrics are likely to be close to zero. We also find that the annual change in GAAP and cash ETRs are similar and that both are increasing on average (0.0188 and 0.0173, respectively). Increasing tax rates are likely to be accompanied by increasing profitability or decreasing costs, or both, confirmed by the average ΔROA (0.0014) and ΔSGA (−0.0051).

Panel B shows deal-related descriptive statistics for significant IPOs. Because we focus on significant IPOs, our sample IPOs receive large proceeds from the offering on average. The mean total assets of our significant IPOs (\$3,485 million) are greater than that of Hsu et al. (2010) (\$1,536 million), which is likely because of our sample selection criterion of dropping loss firms. The mean of total assets for significant IPOs is not statistically different from the mean of the average total assets of the incumbent firm-years associated with each significant IPO (\$4,458 million; untabulated p value for the test of difference is 0.31). Most significant IPOs have Big N auditors and top investment bank certifications.

Our sample uses a rolling window during which we expect an incumbent to observe an IPO's disclosed ETR in year $t - 1$ and to adjust their ETR in year t to imitate the IPO's year $t - 1$ ETR. In Figure 1, we provide a univariate summary of the herding response by incumbent firms in the three years following the significant IPO. We display the mean IPO year $t - 1$ ETR (first bar) and incumbent year $t - 1$ (second bar) ETR. In the third bar, we display incumbent subsequent year t ETR to visualize the average one-year change in incumbent ETRs. Figure 1, (a) and (c), summarizes these means when the incumbent ETR exceeds the IPO ETR in year $t - 1$ for GAAP and cash ETRs, respectively. Figure 1, (b) and (d), summarizes these means when the incumbent ETR is less than the IPO ETR in year $t - 1$ for GAAP and cash ETRs, respectively. Figure 1(a) shows that in each year following the IPO year, the mean GAAP ETR decreases for incumbents with incentives to decrease their GAAP ETR to imitate the significant IPO's GAAP ETR. Figure 1(b) shows that the mean GAAP ETR increases for incumbents with incentives to increase their GAAP ETR. Figure 1, (c) and (d), presents patterns that are similarly consistent with a herding response by incumbent firms.

4. Empirical Results

4.1. Do Industry Incumbents Imitate Tax Policies of Significant IPOs?

We report our tests of Hypothesis 1 and estimates of Equation (1) using GAAP ETRs in column 1 of Table 3. The $Diff_LagETR_IPO$ coefficient is negative and significant (−0.041, $t = -4.68$), suggesting that incumbent firms reduce differences between their GAAP ETRs and

Table 2. Descriptive Statistics

Panel A: Descriptive statistics of the full sample							
Levels variables			Differences/changes variables				
	Mean	Median	Standard deviation		Mean	Median	Standard deviation
CASHETR	0.2651	0.2480	0.1977				
LagCASHETR	0.2478	0.2360	0.1968	Δ CASHETR	0.0173	0.0067	0.2170
LagCASHETR_IPO	0.2637	0.2303	0.2103	Diff_LagETR_IPO (Cash)	−0.0159	−0.0040	0.2870
CASHETR_Median	0.2082	0.2138	0.0721	$\text{Diff_LagETR_Median}$ (Cash)	0.0396	0.0200	0.1928
GAAPETR	0.3211	0.3407	0.1467				
LagGAAPETR	0.3023	0.3380	0.1583	Δ GAAPETR	0.0188	0.0004	0.1709
LagGAAPETR_IPO	0.3159	0.3391	0.1602	Diff_LagETR_IPO (GAAP)	−0.0135	−0.0030	0.2215
GAAPETR_Median	0.3050	0.3202	0.0552	$\text{Diff_LagETR_Median}$ (GAAP)	−0.0027	0.0165	0.1567
RD	0.0228	0.0000	0.0464	Δ RD	−0.0002	0.0000	0.0122
AD	0.0144	0.0000	0.0373	Δ AD	−0.0003	0.0000	0.0086
SGA	0.2622	0.2116	0.2396	Δ SGA	−0.0051	0.0000	0.0462
CAPEXP	0.1196	0.0927	0.0902	Δ CAPEXP	0.0013	0.0017	0.0715
CASH	0.1456	0.0738	0.1736	Δ CASH	−0.0041	−0.0001	0.0833
FI	0.0156	0.0000	0.0333	Δ FI	0.0005	0.0000	0.0161
EQINC	0.0009	0.0000	0.0041	Δ EQINC	0.0000	0.0000	0.0022
NOL	0.3524	0.0000	0.4777	Δ NOL	0.0109	0.0000	0.2494
INTAN	0.1606	0.0778	0.1969	Δ INTAN	0.0089	0.0000	0.0775
PPE	0.3173	0.2322	0.2594	Δ PPE	−0.0017	−0.0025	0.0479
FCF	0.0521	0.0488	0.0919	Δ FCF	−0.0017	−0.0015	0.0871
SIZE	6.3631	6.3939	2.1152	Δ SIZE	0.1225	0.0760	0.2119
ROA	0.1084	0.0874	0.0876	Δ ROA	0.0014	0.0016	0.0661
LEV	0.1868	0.1600	0.1797	Δ LEV	0.0019	0.0000	0.0739
Panel B: Descriptive statistics of significant IPOs							
	Mean	Median	Standard deviation				
GROSSPROCEEDS (\$m)	839.3518	340.0000	1,828.8866				
INITIALRETURN	13.8906	9.3529	16.9744				
UNDERWRITER_RANK	8.7162	9.0000	1.0371				
BIGN	0.9279	1.0000	0.2598				
TOTALASSETS_PreIPO	3,484.8008	894.6870	7,075.8951				
ROA_PreIPO	0.1102	0.0786	0.1179				

Notes. This table presents descriptive statistics for both the levels and changes in our primary regression variables for the full sample in Panel A. Panel B shows descriptive statistics for the 111 significant IPOs. We winsorize all continuous variables except ETRs at the 1st and 99th percentiles for the full sample, and we reset ETRs to fall between zero and one before differencing them. We include detailed variable definitions in the Appendix.

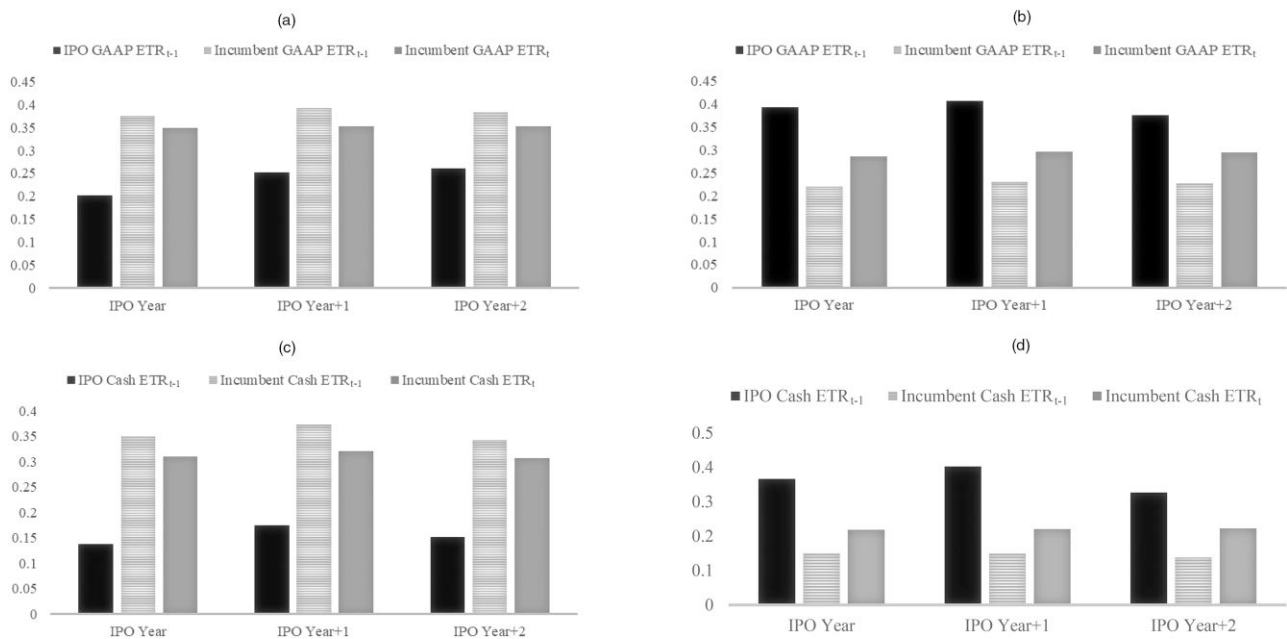
the significant IPO's GAAP ETR in the three years following the IPO. We report results using cash ETRs in column 2 of Table 3. The Diff_LagETR_IPO coefficient is negative and significant (-0.031 , $t = -3.73$). This result suggests that incumbent firms also change their cash ETR in response to the distance between it and the significant IPO's cash ETR in year $t - 1$. This result is consistent with Hypothesis 1 and suggests that incumbent firms imitate a significant IPO's tax outcomes in their industry. In both columns of Table 3, the $\text{Diff_LagETR_Median}$ coefficient is negative and significant, and the magnitude is large. This result is consistent with incumbent firms' tax outcomes responding to an industry median benchmark (Kubick et al. 2015, Kim et al. 2019).¹⁸ Because our variable of interest is significant even after controlling for $\text{Diff_LagETR_Median}$, we interpret our results as providing evidence that the imitation of a significant IPO's tax policies is incremental to

industry-wide median reversion. Overall, we find consistent support for Hypothesis 1 and conclude that industry incumbents exhibit a herding response to significant IPOs' tax policies for up to three years following the IPO year.

4.2. Cross-Sectional and Subsample Analysis

4.2.1. Attention and Information Effects. This section assesses the cross-sectional effects of attention and information on the herding response. We present the results of this analysis in Table 4. Herding behavior is more likely to occur when media attention is high because of its effect on IPO investors' opinions and demands (Liu et al. 2014a), investor familiarity and acceptance of an IPO firm's policies, and investors' risk perceptions (Pollock and Rindova 2003). Furthermore, attention and reputation effects increase the incentive for managers to report ETRs that are comparable to

Figure 1. Mean IPO and Incumbent Year $t - 1$ ETRs and Mean Incumbent Subsequent Year ETR



Notes. This figure presents the mean incumbent and IPO year $t - 1$ ETR and the incumbent year t ETR for the IPO year and the subsequent two years. (a) and (b) present the mean GAAP ETRs for incumbents whose year $t - 1$ GAAP ETR is greater and less than the IPO's year $t - 1$ GAAP ETR, respectively. (c) and (d) present the mean cash ETRs for incumbents whose year $t - 1$ cash ETR is greater and less than the IPO's year $t - 1$ cash ETR, respectively.

peers. Thus, we expect the association between incumbent firms' and significant IPO tax policies to be stronger for significant IPOs that attract the most media attention in the six months preceding their IPO. We test this assertion in columns 1 and 2 of Table 4 by including *IPONEWS_High*, an indicator equal to one when the number of news items listed on LexisNexis in the six months before the IPO date is higher than the median number of news events in the six months before all significant IPOs (zero otherwise). We interact *IPONEWS_High* with *Diff_LagETR_IPO* in our estimation of Equation (1). In column 1, we find a significant negative *Diff_LagETR_IPO* $\times$ *IPONEWS_High* coefficient (-0.049 , $t = -1.96$), suggesting incumbent firms' GAAP ETRs respond to more visible significant IPOs. However, in column 2, the *Diff_LagETR_IPO* $\times$ *IPONEWS_High* coefficient is insignificant (-0.016 , $t = -0.63$). Therefore, we suggest that the visibility of a significant IPO elicits managers to adjust their more visible GAAP ETRs but not their less visible cash ETRs.

Under herding theory, other firms will imitate the most readily available information and socially acceptable focal firm attributes to reduce uncertainty and gain favor (Pollock et al. 2008). Lost investor attention also likely captures aspects of reputational risk that accompany managers' tax policy choices. As a result, we also expect herding behavior to be more pronounced in the subset of firms that lose investor attention in the month of the significant IPO. We test this expectation in columns 3 and 4 of Table 4 using the Google trends measure of investor attention introduced by

Da et al. (2011). We include an indicator variable, *IncumbentAttention_Decrease*, equal to one if the incumbent experienced a reduction in investor attention in the month of the significant IPO and zero otherwise, and its interaction with *Diff_LagETR_IPO* in our estimation of Equation (1). In column 3, we find a significant negative *Diff_LagETR_IPO* $\times$ *IncumbentAttention_Decrease* coefficient (-0.042 , $t = -1.86$), suggesting incumbent firms' GAAP ETRs respond more to significant IPOs when they have recently experienced a decrease in investor attention. However, in column 4, the *Diff_LagETR_IPO* $\times$ *IncumbentAttention_Decrease* coefficient is insignificant (-0.003 , $t = -0.13$), further supporting the visibility of GAAP ETRs relative to cash ETRs.

When motivating our hypotheses, we assert that the IPO setting is characterized by uncertainty because incumbent firms and their investors lack publicly available information against which to compare the IPO firm's policy choices prior to the IPO. However, for IPO firms recently publicly traded, incumbent firms and their investors have some publicly available financial statement information against which to compare their policy choices. Thus, public IPOs going dark and entering our sample as a significant IPO upon their second IPO could be associated with less uncertainty and be less of a shock to industry norms. We test this expectation in columns 5 and 6 of Table 4. We include an indicator variable, *Recent_Public*, equal to one if the significant IPO was a public company within five years of the current IPO (zero otherwise), and its interaction with *Diff_LagETR_IPO* in our estimation of Equation (1). The

Table 3. Incumbent Tax Policy Response in the Three Years Following the Significant IPO

Variables	$\Delta GAAPETR$ (1)	$\Delta CASHETR$ (2)
<i>Diff_LagETR_IPO</i>	−0.041*** (−4.68)	−0.031*** (−3.73)
<i>Diff_LagETR_Median</i>	−0.592*** (−34.23)	−0.583*** (−34.14)
ΔRD	−0.005 (−0.05)	−0.087 (−0.58)
ΔAD	−0.061 (−0.42)	−0.320 (−1.50)
ΔSGA	−0.019 (−0.56)	0.112** (2.57)
$\Delta CAPEXP$	−0.042* (−1.87)	−0.024 (−0.77)
$\Delta CASH$	0.007 (0.33)	−0.075*** (−2.77)
ΔFI	−0.384*** (−2.88)	−0.594*** (−4.36)
$\Delta EQINC$	−1.496** (−2.08)	−1.501 (−1.46)
ΔNOL	−0.002 (−0.35)	0.003 (0.44)
$\Delta INTAN$	0.016 (0.69)	−0.124*** (−4.90)
ΔPPE	0.091*** (2.71)	−0.154*** (−3.60)
ΔFCF	−0.053*** (−2.64)	−0.211*** (−7.61)
$\Delta SIZE$	0.046*** (5.10)	−0.029*** (−2.74)
ΔROA	0.322*** (9.60)	−0.400*** (−10.70)
ΔLEV	−0.006 (−0.26)	0.065** (2.53)
Observations	12,211	12,211
Adjusted R^2	0.375	0.345
Year fixed effects	Yes	Yes

Notes. This table presents the results of estimating Equation (1) using $\Delta GAAPETR$ in column 1 and $\Delta CASHETR$ in column 2. We do not report the intercept as all specifications include year-fixed effects. We winsorize all continuous variables except ETRs at the 1st and 99th percentiles. We reset ETRs to fall between zero and one before differencing them. We report t statistics with robust standard errors clustered by firm in parentheses. We include detailed variable definitions in the Appendix.

*, **, and ***Significance at the 0.10, 0.05, and 0.01 levels using a two-tailed test, respectively.

estimated $Diff_LagETR_IPO \times Recent_Public$ coefficient is insignificant in either column 5 or column 6. An F test on the sum of $Diff_LagETR_IPO \times Recent_Public$ and $Diff_LagETR_IPO$ suggests we cannot reject the null of no herding response for incumbents matched to a significant IPO with a recent history of being publicly traded ($p = 0.7482$ for GAAP ETR; $p = 0.2906$ for cash ETR). This result suggests that herding does not occur when the IPO reveals less information.

4.2.2. Tax-Specific Attention Effects. In this section, we examine the effect of investor attention to taxes on the herding responses of incumbent firms. Incumbent

firms with analysts that provide “implicit ETR forecasts” (through the issuance of both pretax and after-tax EPS forecasts) should represent incumbent investor attentiveness to taxes because analysts reflect the information demanded by investors (DeFond and Hung 2007, Barniv and Cao 2009, Barniv et al. 2010, Amiram et al. 2018, Mauler 2019). Among the incumbents with analyst forecasts, the herding response of incumbent firms toward significant IPOs’ tax policies should be stronger for incumbents with analysts who provide implicit ETR forecasts. We test this assertion by including in our estimation of Equation (1) *ImplicitETR_FC*, an indicator variable equal to one for incumbents for whom at least one analyst provides an implicit ETR forecast consistent with Bratten et al. (2017) and zero for all other incumbent firms included in the I/B/E/S database. We also include the interaction of *ImplicitETR_FC* with *Diff_LagETR_IPO*. This cross-sectional analysis has substantially fewer firm years because not all incumbent firms have analyst coverage. Despite the smaller sample size, in column 1 of Table 5, we find a significant negative $Diff_LagETR_IPO \times ImplicitETR_FC$ coefficient (-0.037 , $t = -1.68$), suggesting that incumbent firms’ GAAP ETR responses are more pronounced when investors are likely to care more about incumbents’ taxes. In column 2, the $Diff_LagETR_IPO \times ImplicitETR_FC$ coefficient is insignificant (-0.001 , $t = -0.08$). Although analysts will often exclude certain transitory items when forecasting EPS, implicit ETR forecasts use GAAP EPS, which likely explains why GAAP, but not cash ETRs, respond to investors’ attention to taxes.

Because taxes are likely more salient to investors in periods associated with significant changes in corporate tax laws, herding should be more pronounced in these periods. To test this assertion in Table 5, we include in our estimation of Equation (1) the interactions between *Diff_LagETR_IPO* and a series of indicator variables that represent the periods in our sample associated with significant corporate tax changes. These include the Job Creation and Worker Assistance Act of 2002, the Jobs and Growth Tax Relief Reconciliation Act of 2003, the American Jobs Creation Act of 2004 (2002_2004); the American Recovery and Reinvestment Tax Act (2009); the American Taxpayer Relief Act of 2012 (2012); and the Tax Cuts and Jobs Act (2017). For both cash and GAAP ETR, we find that the *Diff_LagETR_IPO* coefficient continues to be negative and significant. We also find evidence consistent with a stronger herding response of incumbent GAAP (cash) ETRs for three (one) of the four periods when we expect corporate taxes to be more salient for investors.

4.2.3. Costs of Tax Expense Changes. We next assess whether the incumbent herding response varies with the costs of tax expense changes. Corporate tax outcomes

Table 4. Cross-Sectional Effects of Information and Attention

Variables	IPO media attention		Incumbent investor attention		Previously public IPOs	
	Δ GAAPETR (1)	Δ CASHETR (2)	Δ GAAPETR (3)	Δ CASHETR (4)	Δ GAAPETR (5)	Δ CASHETR (6)
<i>Diff_LagETR_IPO</i>	0.009 (0.41)	−0.016 (−0.69)	−0.013 (−0.80)	−0.010 (−0.63)	−0.039*** (−4.45)	−0.032*** (−2.95)
<i>IPONEWS_High</i>	0.019*** (4.85)	0.004 (0.83)	—	—	—	—
<i>Diff_LagETR_IPO</i> × <i>IPONEWS_High</i>	−0.049** (−1.96)	−0.016 (−0.63)	—	—	—	—
<i>IncumbentAttention_Decrease</i>	—	—	0.006 (1.49)	0.003 (0.62)	—	—
<i>Diff_LagETR_IPO</i> × <i>IncumbentAttention_Decrease</i>	—	—	−0.042* (−1.86)	−0.003 (−0.13)	—	—
<i>Recent_Public</i>	—	—	—	—	−0.017*** (−4.38)	0.003 (0.30)
<i>Diff_LagETR_IPO</i> × <i>Recent_Public</i>	—	—	—	—	0.026 (0.63)	0.007 (0.26)
<i>F</i> test: <i>Diff_LagETR_IPO</i> + <i>Diff_LagETR_IPO</i> × <i>Recent_Public</i>	—	—	—	—	0.10 (<i>p</i> = 0.7482)	1.12 (<i>p</i> = 0.2906)
Observations	7,828	7,828	5,272	5,272	12,211	12,211
Adjusted <i>R</i> ²	0.378	0.347	0.400	0.332	0.376	0.345
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes. In columns 1 and 2, we include in our estimation *IPONEWS_High* equal to one for significant IPOs with above-median media coverage in the six months preceding the IPO. In columns 3 and 4, we include an indicator variable if the incumbent's Google Trends score decreased in the month of the significant IPO (*IncumbentAttention_Decrease*). In columns 5 and 6, we estimate Equation (1), including an indicator variable for incumbents associated with IPO firms that were previously public within five years of the IPO year in our sample (*Recent_Public*). Control variables are unreported for brevity. We winsorize all continuous variables except ETRs at the 1st and 99th percentiles, and we reset ETRs to fall between zero and one before differencing them. We report *t* statistics with robust standard errors clustered by firm in parentheses. We include detailed variable definitions in the Appendix.

*, **, and ***Significance at the 0.10, 0.05, and 0.01 levels using a two-tailed test, respectively.

represent an equilibrium that balances the nontax costs with the benefits of tax planning (Kim et al. 2019). Tax planning is not costless; high nontax costs accompany substantial movements in tax outcomes. These costs could include contracting costs to unwind existing tax planning or professional service fees to implement new tax planning. There are also limits to firms' tax planning opportunities and tax-saving strategies because of implementation and timing issues (e.g., transfer pricing strategies) or because firms have exhausted available tax planning opportunities. Tax policy choices, such as those that involve organizational structure changes, can also be challenging to implement and costly to abandon. We expect the herding response to be stronger when significant IPOs' tax policies are initially closer to incumbents' tax policies. This closeness provides incumbents with the need for smaller adjustments to their ETR to herd relative to incumbents that require larger ETR changes to achieve their herding objective.

We test this assertion by including an indicator variable, *Same_Quint*, equal to one if the incumbent's year *t* − 1 ETR is in the same quintile as the IPO's year *t* − 1 ETR and its interaction with *Diff_LagETR_IPO*. In columns 1 and 2 of Table 6, Panel A, we document a significant negative main effect for both cash ETR and GAAP

ETR (−0.035, *t* = −3.90 for GAAP ETR; −0.026, *t* = −3.04 for cash ETR), indicating that incumbents that require larger ETR changes to herd exhibit a herding response. However, the estimated *Diff_LagETR_IPO* × *Same_Quint* coefficient is also negative and significant (−0.142, *t* = −3.88 for GAAP ETR; −0.139, *t* = −3.55 for cash ETR), indicating a stronger herding response by incumbents for which only small ETR changes achieve the herding objective. Our results are consistent with the notion that small ETR changes are less costly than large ETR changes.

In Panel B of Table 6, we also condition incumbent firm observations on whether their year *t* − 1 ETR is above or below the IPO's ETR in *t* − 1 and include *Same_Quint* and its interaction with *Diff_LagETR_IPO*. The most significant responses are found within the firms herding "down" toward the significant IPO. The results in Panel B of Table 6 also suggest that the cost of small ETR changes is lower when the incumbent must decrease its ETR to herd toward the IPO. Reputational costs are increasing in tax avoidance. Small (large) reductions in the ETR are associated with relatively smaller (larger) costs. There are also explicit costs to tax planning; the implementation costs of small ETR reductions are likely lower than implementation costs

Table 5. Cross-Sectional Effects of Tax-Specific Attention

Variables	Incumbent with analyst implied ETR forecast		Enactment years of major corporate tax legislation	
	$\Delta GAAPETR$ (1)	$\Delta CASHETR$ (2)	$\Delta GAAPETR$ (3)	$\Delta CASHETR$ (4)
<i>Diff_LagETR_IPO</i>	−0.031 (−1.57)	−0.023 (−1.58)	−0.024*** (−2.58)	−0.052*** (−5.81)
<i>ImplicitETR_FC</i>	−0.004 (−1.18)	−0.019*** (−3.94)	—	—
<i>Diff_LagETR_IPO</i> × <i>ImplicitETR_FC</i>	−0.037* (−1.68)	−0.001 (−0.08)	—	—
<i>2002_2004</i>	—	—	−0.008** (−2.36)	−0.023*** (−4.69)
<i>2009</i>	—	—	0.022*** (2.67)	−0.007 (−0.76)
<i>2012</i>	—	—	0.005 (0.94)	0.017*** (2.71)
<i>2017</i>	—	—	0.000 (0.00)	−0.032** (−2.21)
<i>Diff_LagETR_IPO</i> × <i>2002_2004</i>	—	—	−0.066*** (−2.60)	−0.020 (−1.05)
<i>Diff_LagETR_IPO</i> × <i>2009</i>	—	—	0.020 (0.45)	−0.056 (−1.20)
<i>Diff_LagETR_IPO</i> × <i>2012</i>	—	—	−0.087** (−2.38)	0.106** (2.32)
<i>Diff_LagETR_IPO</i> × <i>2017</i>	—	—	−0.299* (−1.95)	−0.018 (−0.12)
Observations	7,637	7,637	12,211	12,211
Adjusted R^2	0.379	0.331	0.372	0.337
Controls	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	No	No

Notes. This table presents the results of cross-sectional tests of the effect of tax-specific attention on the herding response of incumbents. In columns 1 and 2, we include an indicator variable equal to one if an analyst issues an implicit ETR forecast for an incumbent firm (*ImplicitETR_FC*) in year t , where an implicit ETR forecast is the presence of both pre- and after-tax earnings forecasts by an analyst. In columns 3 and 4, we include indicator variables equal to one for years in which major corporate tax legislation was passed: *2002_2004* is one for the enactments of the Job Creation and Worker Assistance Act of 2002, the Jobs and Growth Tax Relief Reconciliation Act of 2003, and the American Jobs Creation Act of 2004. *2009* is equal to one due to the enactment of the American Recovery and Reinvestment Tax Act of the American Recovery and Reinvestment Act of 2009. *2012* is equal to one due to the enactment of the American Taxpayer Relief Act of 2012. *2017* is equal to one due to the enactment of the Tax Cuts and Jobs Act of 2017. We winsorize all continuous variables except ETRs at the 1st and 99th percentiles, and we reset ETRs to fall between zero and one before differencing them. We report t statistics with robust standard errors clustered by firm in parentheses. We include detailed variable definitions in the Appendix.

*, **, and ***Significance at the 0.10, 0.05, and 0.01 levels using a two-tailed test, respectively.

for large ETR reductions. Therefore, this result is consistent with a stronger herding response to decreasing the GAAP ETR when nontax costs to move the ETR are relatively low and do not outweigh the benefits of increased investor and analyst attention.

The cost of reporting an ETR lower than the significant IPO's GAAP or cash ETR is likely greater for an incumbent firm that is potentially viewed, *ex ante*, as a tax-avoiding firm. If an incumbent firm's ETR is low relative to an industry standard (such as the industry median), then the attention directed toward a higher significant IPO's ETR could warrant additional scrutiny and increased reputational cost for the tax-avoiding incumbent firm. We therefore expect a stronger upward herding response for incumbents whose year $t - 1$ ETR is below the industry median. In Panel C of Table 6, we include in the estimation of Equation 1 an indicator variable, *Low_ETR*, equal to one if the incumbent's year $t - 1$

ETR is below the industry median and interact it with *Diff_LagETR_IPO*.¹⁹ For this analysis, we focus on the subset of incumbent firm-years whose year $t - 1$ ETR is less than the significant IPO's ETR.

Consistent with expectations, we find a significantly greater upward GAAP ETR herding response among incumbents with low year $t - 1$ GAAP ETRs. We continue to document a statistically significant upward GAAP ETR herding response for incumbents that do not appear, by our classification at least, to be tax avoiders. However, estimating the economic magnitude of the upward herding response for low ETR incumbents versus all other incumbents implies an average upward herding response of five percentage points for tax-avoiding firms versus 0.6 percentage points for nontax avoiding firms.²⁰ This result suggests that the upward herding response for nonlow ETR incumbents is not economically meaningful. Thus, consistent with

Table 6. Cross-Sectional Effects of the Costs of Tax Expense Changes

Panel A: Distance from the IPO								
	$\Delta GAAPETR$				$\Delta CASHETR$			
	(1)				(2)			
$Diff_LagETR_IPO$	−0.035*** (−3.90)				−0.026*** (−3.04)			
$Same_Quint$	−0.003 (−0.95)				−0.001 (−0.22)			
$Diff_LagETR_IPO \times Same_Quint$	−0.142*** (−3.88)				−0.139*** (−3.55)			
Observations	12,211				12,211			
Adjusted R^2	0.377				0.347			
Controls	Yes				Yes			
Year fixed effects	Yes				Yes			
Panel B: Distance from the IPO for incumbents whose ETR is above/below the IPO								
	$\Delta GAAPETR$				$\Delta CASHETR$			
	Higher than IPO		Lower than IPO		Higher than IPO		Lower than IPO	
	Same quintile (1)	Different quintile (2)	Same quintile (3)	Different quintile (4)	Same quintile (5)	Different quintile (6)	Same quintile (7)	Different quintile (8)
$Diff_LagETR_IPO$	−0.296*** (−3.61)	−0.093*** (−5.49)	−0.135*** (−2.66)	−0.077*** (−3.44)	−0.477*** (−4.90)	−0.082*** (−2.75)	0.077 (1.25)	−0.038*** (−2.61)
Same versus different		0.203**		0.058		0.395***		−0.115*
p value		0.0140		0.2864		0.0001		0.0650
Observations	1,278	4,639	1,578	4,716	1,401	4,539	1,298	4,973
Adjusted R^2	0.466	0.348	0.306	0.281	0.458	0.414	0.194	0.096
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Panel C: Effect of incumbent tax avoidance on upward ETR herding								
	Lower than IPO							
	$\Delta GAAPETR$				$\Delta CASHETR$			
	(1)				(2)			
$Diff_LagETR_IPO$	−0.042** (−2.13)				−0.030 (−1.45)			
Low_ETR	0.053*** (10.11)				0.083*** (12.49)			
$Diff_LagETR_IPO \times Low_ETR$	−0.271*** (−9.85)				−0.036 (−1.43)			
F test: $Diff_LagETR_IPO + Diff_LagETR_IPO \times Low_ETR$	−0.313*** ($p = 0.0000$)				−0.066*** ($p = 0.0000$)			
Observations	6,294				6,271			
Adjusted R^2	0.216				0.099			
Controls	Yes				Yes			
Year fixed effects	Yes				Yes			

Notes. This table presents the cross-sectional effects of the costs of tax expense changes on the herding response of incumbents. In Panel A, we include an indicator variable, *Same_Quint*, equal to one when the incumbent firm's year $t - 1$ ETR is in the same quintile as the IPO firm's year $t - 1$ ETR. In Panel B, we partition incumbent firm observations first on whether their year $t - 1$ ETR is above or below the IPO's ETR in $t - 1$ and second on whether it was in the same quintile as the IPOs. We do not report the intercept as all specifications include year-fixed effects. In Panel C, we include an indicator variable, *Low_ETR*, equal to one if the incumbent's year $t - 1$ ETR is below the industry median. In our Panel C estimations, we use the subsample of incumbent firm-years whose year $t - 1$ ETR is less than the significant IPO's year $t - 1$ ETR and exclude the distance between the incumbent's year $t - 1$ ETR and the industry median. We do not report control variables for brevity. We winsorize all continuous variables except ETRs at the 1st and 99th percentiles, and we reset ETRs to fall between zero and one before differencing them. We report *t* statistics with robust standard errors clustered by firm in parentheses. We include detailed variable definitions in the Appendix.

*, **, and ***Significance at the 0.10, 0.05, and 0.01 levels using a two-tailed test, respectively.

expectations, we find a statistically and economically larger herding response for low ETR (i.e., “tax-avoiding”) incumbents. Concerning cash ETR, our

results summarized in column 2 of Table 6, Panel C indicate no evidence of an upward herding response for incumbents that are not tax avoiders. However, a

statistically significant F test on the sum of the coefficients on $Diff_LagETR_IPO$ and the interaction between $Diff_LagETR_IPO$ and Low_ETR in column 2 suggest that the upward cash ETR herding response is statistically significant for tax-avoiding incumbents.

4.3. Consequences of Herding

Our cross-sectional analyses reveal results consistent with the ex ante costs and benefits of imitating a significant IPO's tax policies. In this section, we also examine the ex post benefits of herding. Our hypothesis implies that investor attention on the IPO, and incumbents' desire to maintain investor attention, explains imitation of IPO tax policy. To substantiate this assertion, we examine whether incumbents' imitation of IPOs' tax outcomes is associated with market-based attention

outcomes, including increased analyst following and increased investor attention using Google Trends scores (Da et al. 2011). We regress these market-based attention measures on a set of controls consistent with prior research and indicator variables (i.e., $Herd_CashETR$ or $Herd_GAAPETR$) that equal one for the subset of incumbents with the strongest evidence of a herding response. Specifically, $Herd_CashETR$ and $Herd_GAAPETR$ are equal to one if the incumbent firm is in the bottom quartile of the product of $\Delta CashETR \times Diff_LagETR_IPO$, or $\Delta GAAPETR \times Diff_LagETR_IPO$, and zero if in the top quartile. This specification compares firms with the most movement toward the UPO's ETR (i.e., more negative values) to firms with the least movement toward the IPO's ETR. We present the results of these tests in Table 7. We include variable definitions for each test in the Appendix.

Table 7. Consequences of Herding

Variables	Investor attention in 10-K filing month		Change in analyst following	
	(1)	(2)	(3)	(4)
$Herd_GAAPETR$	0.038** (2.10)	—	0.022* (1.71)	—
$Herd_CashETR$	—	−0.022 (−1.17)	—	0.027** (2.00)
$ShareTurnover$	0.162*** (2.62)	0.170*** (3.02)	—	—
$AnalystFollowing$	−0.061*** (−3.83)	−0.043*** (−2.73)	—	—
MVE	0.044*** (5.65)	0.032*** (3.94)	—	—
MTB	−0.000 (−0.96)	−0.000 (−0.38)	—	—
$Attention_PrevYear$	1.025*** (51.82)	0.980*** (49.62)	—	—
$BHAR_10kmonth$	0.309* (1.96)	0.063 (0.40)	—	—
$\Delta SIZE$	—	—	0.063* (1.94)	0.124*** (3.15)
$\Delta INTAN$	—	—	0.321*** (3.87)	0.373*** (3.75)
ΔMTB	—	—	0.095*** (7.07)	0.089*** (6.71)
$\Delta SecurityIssuance$	—	—	−0.015 (−0.78)	−0.024 (−1.08)
$\Delta Return$	—	—	−0.027* (−1.93)	−0.014 (−0.99)
$\Delta ShareTurnover$	—	—	0.290*** (2.82)	0.565*** (4.79)
$\Delta Return_Volatility$	—	—	−0.111 (−0.62)	−0.511* (−1.84)
Observations	1,765	1,765	3,771	3,771
Adjusted R^2	0.658	0.616	0.042	0.051
Year fixed effects	Yes	Yes	Yes	Yes

Notes. We regress a specific incumbent outcome variable on an indicator ($Herd_CashETR$, $Herd_GAAPETR$) equal to one when the incumbent firm is in the bottom quartile of the product of $\Delta Cash(GAAP)ETR \times Diff_LagETR_IPO$, zero if in the top quartile. In columns 1 and 2, our outcome variable (*Investor Attention*) is the natural logarithm of Google Trends score in the month of the incumbent 10-K filing. In columns 3 and 4, our outcome variable (*%Change Analyst Following*) is the percentage change in analyst following from year t to year $t + 1$. We define control variables in the Appendix. We report t statistics with robust standard errors clustered by firm in parentheses.

*, **, and ***Significance at the 0.10, 0.05, and 0.01 levels using a two-tailed test, respectively.

Column 1 of Table 7 indicates that incumbents who herd their GAAP ETR toward the IPO firm's GAAP ETR experience greater investor attention in the month of their 10-K filing than nonherding incumbents. This result is consistent with herding being an effective strategy to regain investor attention following the IPO. Given that increased investor attention is associated with higher stock prices in the two weeks following the increase in attention (Da et al. 2011), this indicates a significant consequence to incumbent firms. We find in columns 3 and 4 that incumbents that herd toward the IPO's cash or GAAP ETR have a greater analyst following as a reward for herding. The association between analyst following and increased market liquidity reduces a firm's cost of capital (Roulstone 2003); thus, a larger increase in analyst following for herding incumbents represents a benefit to them. Overall, we find that herding firms experience larger positive capital market outcomes than nonherding firms.

4.4. How Industry Incumbents Imitate

4.4.1. Tax Footnote Disclosure. As an alternative means to assess a tax-specific herding response, we examine whether the tax disclosures of incumbents become more similar to the tax disclosures of the significant IPO firms following the IPO. Disclosure similarity is measured as the cosine similarity of tax footnotes consistent with recent disclosure research (Dyer et al. 2020, Glaeser and Landsman 2021) and obtained from SeekEdgar. We then model the change in cosine similarity of incumbents' tax footnote disclosures from year $t - 1$ to year t with the significant IPO's tax footnote disclosures in year $t - 1$ as a function of *Herd_CashETR* or *Herd_GAAPETR* (as defined previously) and control variables used in our primary model. We tabulate the results of this analysis in Panel A of Table 8.

For incumbents with the most observable herding response to a significant IPO's GAAP ETR, we find in columns 2 and 3 that their 10-K tax footnote disclosures become more similar to significant IPOs' tax footnotes after the IPO. This similarity exists even after controlling for changes in nontax footnote similarity (i.e., column 3). The significant result for GAAP ETR but not cash ETR (i.e., columns 5 and 6) could be partly the result of the GAAP ETR's relatively more prominent visibility in the 10-K. Firms' financial statements typically do not include separate cash ETR disclosures, and cash taxes paid are a line item in the Statement of Cash Flows. Importantly, we also examine whether the *nontax* disclosures of incumbent firms who herd their GAAP and cash ETR change following the IPO (i.e., columns 1 and 4). We find that while tax footnote disclosures become more similar following the IPO, nontax disclosures do not. This analysis underscores

the importance of herding tax disclosures following a significant IPO in an industry. Furthermore, this analysis helps address the concern that our observed results are due to other corporate policy changes that lead to secondary tax effects.

4.4.2. ETR Increases vs. Decreases. In Panel B of Table 8, we partition our primary sample on whether the incumbent firm's ETR is greater or less than the significant IPO's ETR and estimate Equation (1) separately for each subsample. The estimated *Diff_LagETR_IPO* coefficients are negative and significant for firms whose ETR is above or below the significant IPO's ETR for GAAP and cash ETR measures. Therefore, we find that incumbent firms imitate a significant IPO by increasing and decreasing their cash and GAAP ETRs depending on whether their ETR is above or below the significant IPO's ETR. *F* tests reveal that the estimated coefficients for incumbent firms with ETRs greater (less) than a significant IPO's ETR are not significantly different ($\Delta\text{GAAPETR}$: $p = 0.3127$; $\Delta\text{CASHETR}$: $p = 0.2475$). This result supports the notion that incumbent firms exhibit symmetric responses in tax policy changes consistent with herding motives and less consistent with the rivalry motives that yield only ETR decreases in other studies of peer tax effects.

4.4.3. Tax Accruals vs. Cash Tax Planning. We consistently find that incumbent firms imitate both the cash and GAAP ETRs of significant IPOs. Incumbent firms' cash ETR imitation response is nearly identical to the GAAP ETR response for the full sample. However, cross-sectional analyses reveal variation in the imitation response across the two measures, where cash ETRs sometimes respond to a lesser degree than GAAP ETRs in different settings. In this section, we examine whether these differences result from the mechanisms employed by incumbent firms to imitate significant IPO tax policies.

Firm management has several means of imitating a significant IPO's tax policies. These include (1) changes in tax policies with immediate changes to cash taxes paid, (2) changes in tax policy with future changes to cash taxes paid that also affect current period financial statement tax accruals, or (3) adjusting tax accruals that do not affect cash taxes paid (e.g., adjustments to the deferred tax valuation allowance or uncertain tax benefit reserves). An incumbent firm wishing to move its cash ETR toward a significant IPO cash ETR benchmark can only do so through tax planning changes that affect current cash flows. However, an incumbent firm wishing to move its GAAP ETR toward a significant IPO benchmark can theoretically do so through tax accruals without altering its current cash flows. Thus, *ceteris paribus*, GAAP ETR management is relatively less costly than cash ETR management because firms

Table 8. How Industry Incumbents Herd

Panel A: Tax footnote similarity between herding incumbents and significant IPOs						
	Δ Non-Tax FN Cosine Similarity (1)	Δ Tax FN Cosine Similarity (2)	Δ Tax FN Cosine Similarity (3)	Δ Non-Tax FN Cosine Similarity (4)	Δ Tax FN Cosine Similarity (5)	Δ Tax FN Cosine Similarity (6)
<i>Herd_GAAPETR</i>	−0.000 (−0.09)	0.003* (1.84)	0.003* (1.88)	— —	— —	— —
<i>Herd_CashETR</i>	— —	— —	— —	0.000 (0.00)	−0.000 (−0.24)	−0.000 (−0.24)
Δ Non-Tax FN Cosine Similarity	— —	— —	0.328*** (2.89)	— —	— —	0.427*** (3.02)
Observations	4,569	4,569	4,569	4,569	4,569	4,569
Adjusted R^2	0.011	0.024	0.052	0.009	0.022	0.067
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Panel B: Incumbent ETRs above and below the significant IPOs						
	Higher than IPO Δ GAAPETR (1)	Lower than IPO Δ GAAPETR (2)	Higher than IPO Δ CASHETR (3)	Lower than IPO Δ CASHETR (4)		
<i>Diff_LagETR_IPO</i>	−0.076*** (−4.52)	−0.051*** (−2.70)	−0.064*** (−2.60)	−0.032** (−2.51)		
Above versus below IPO ETR		0.025		0.032		
p value		0.3127		0.2475		
Observations	5,917	6,294	5,940	6,271		
Adjusted R^2	0.374	0.287	0.423	0.125		
Controls	Yes	Yes	Yes	Yes		
Year fixed effects	Yes	Yes	Yes	Yes		
Panel C: Immediate ETR response in the year following the significant IPO						
	Δ GAAPETR (1)		Δ CASHETR (2)			
<i>Diff_LagETR_IPO</i>	−0.046*** (−3.09)		−0.012 (−0.72)			
Observations	4,497		4,497			
Adjusted R^2	0.366		0.356			
Controls	Yes		Yes			
Year fixed effects	Yes		Yes			
Panel D: Abnormal changes in the valuation allowance and reserve for uncertain tax positions						
	Dependent variable: Δ GAAPETR					
	Higher than IPO (1)	Lower than IPO (2)	Higher than IPO (3)	Lower than IPO (4)		
<i>Diff_LagETR_IPO</i>	−0.014 (−0.35)	−0.045 (−1.58)	−0.066** (−2.50)	−0.014 (−0.64)		
<i>Discr_VADecr</i>	0.002 (0.23)	— —	— —	— —		
<i>Diff_LagETR_IPO</i> $\times$ <i>Discr_VADecr</i>	−0.134** (−2.23)	— —	— —	— —		
<i>Discr_VAIncr</i>	— —	0.002 (0.24)	— —	— —		
<i>Diff_LagETR_IPO</i> $\times$ <i>Discr_VAIncr</i>	— —	−0.079* (−1.79)	— —	— —		
<i>Discr_TaxResDecr</i>	— —	— —	−0.012** (−2.11)	— —		
<i>Diff_LagETR_IPO</i> $\times$ <i>Discr_TaxResDecr</i>	— —	— —	−0.029 (−0.76)	— —		
<i>Discr_TaxResIncr</i>	— —	— —	— —	0.003 (0.58)		

Table 8. (Continued)

Panel D: Abnormal changes in the valuation allowance and reserve for uncertain tax positions				
	Dependent variable: $\Delta GAAPETR$			
	Higher than IPO (1)	Lower than IPO (2)	Higher than IPO (3)	Lower than IPO (4)
$Diff_LagETR_IPO \times Discr_TaxResIncr$	—	—	—	−0.076*** (−2.70)
Observations	2,143	2,340	4,841	5,365
Adjusted R^2	0.398	0.345	0.374	0.314
Controls	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes

Notes. This table examines the mechanisms of ETR herding. In Panel A, we regress the year $t - 1$ to year t change in the cosine similarity of an incumbent's tax footnote disclosures to the significant IPO's year $t - 1$ tax footnote disclosures on $Herd_CashETR$ and $Herd_GAAPETR$. Panel B partitions the sample on whether the incumbent's year $t - 1$ ETR was greater or less than the IPO year $t - 1$ ETR. Panel C presents the results of estimating Equation (1) using $\Delta CASHETR$ in column 1 and $\Delta GAAPETR$ in column 2 for only the incumbent year ending in the significant IPO year. Panel D presents the results of testing whether firms' adjustments to their GAAP ETRs occur through discretionary changes in the deferred tax valuation allowance or reserve for uncertain tax positions. Columns 1 and 3 (columns 2 and 4) are the subsamples of incumbent firms whose GAAP ETR in year $t - 1$ is higher (lower) than the significant IPO firm's GAAP ETR in year $t - 1$. $\Delta GAAPETR$ is the dependent variable in all estimations. $Discr_VADecr$ and $Discr_VAIncr$ are indicators of discretionary decreases and increases, respectively, in the deferred tax valuation allowance. $Discr_TaxResDecr$ and $Discr_TaxResIncr$ are indicators of discretionary decreases and increases, respectively, in the reserve for uncertain tax positions. All specifications include year-fixed effects and control variables that are unreported for brevity. We winsorize all continuous variables except ETRs at the 1st and 99th percentiles, and we reset ETRs to fall between zero and one before differencing them. We report t statistics with robust standard errors clustered by firm in parentheses. We include detailed variable definitions in the Appendix.

*, **, and ***Significance at the 0.10, 0.05, and 0.01 levels using a two-tailed test, respectively.

can adjust tax accruals, yielding immediate effects (Dhaliwal et al. 2004).²¹

We empirically evaluate the timing of incumbent firms' imitation responses and whether observed responses result from abnormal adjustments to specific tax accruals. Panel C of Table 8 presents incumbent firms' immediate ETR responses by estimating Equation (1) for only the significant IPO initial offering year. In the IPO year, incumbent firms can observe the significant IPO's ETR in the year preceding the IPO year and adjust their current year's ETR toward that of the significant IPO. If firms can implement tax planning that generates an immediate change in cash taxes, we would observe a significant association between $Diff_LagETR_IPO$ and changes in cash and GAAP ETRs. However, we document a significant negative $Diff_LagETR_IPO$ coefficient when using GAAP ETR changes as the dependent variable but an insignificant coefficient when using cash ETR changes as the dependent variable. These results are consistent with incumbent firms' immediate response through tax accrual adjustments.²²

The deferred tax valuation allowance and reserve for uncertain tax positions represent two specific accrual mechanisms that incumbent firm managers can use to adjust their GAAP ETR up or down toward a significant IPO firm. Thus, we examine whether discretionary changes in each of these accounts are consistent with their use to achieve a herding objective. We obtain changes in the deferred tax valuation allowance from S&P Capital IQ, Audit Analytics, and hand collection.

We measure changes in the reserve for uncertain tax positions following Blouin and Tuna (2007) because separate and uniform disclosure did not occur until 2008. We measure discretionary changes in each specific tax accrual by estimating industry-specific coefficients for each using determinants models from prior literature (Frank and Rego 2006, Gupta et al. 2016). We supplement our valuation allowance determinants model to control for an additional future year of changes in pretax income.²³ Doing so better isolates nondiscretionary accruals and reduces the chances that our results, and the effect of these decisions on GAAP ETR, simply reflect managers' expectations about future performance related to the decision to book a new valuation allowance or increase the existing valuation allowance. We then measure the firm-specific discretionary change as the residual from applying the industry-specific coefficients to firm-specific information (see the Appendix).

Panel D of Table 8 includes indicator variables for discretionary decreases in the deferred tax valuation allowance ($Discr_VADecr$) and reserve for uncertain tax positions ($Discr_TaxResDecr$) when the incumbent firms' ETRs exceed the significant IPOs and separately when the incumbent firms' ETRs are less than the significant IPOs. We examine only $\Delta GAAPETR$ in Panel D of Table 8 because we are interested in how managers imitate a GAAP ETR through tax accruals. We find in column 2 that the estimated $Diff_LagETR \times Discr_VAIncr$ coefficient is negative and significant, as is the estimated $Diff_LagETR \times Discr_TaxResIncr$ coefficient in column 4. We also find in column 1 that the estimated

$Diff_LagETR \times Discr_VADecr$ coefficient is negative and significant. However, we find in column 3 that the estimated $Diff_LagETR \times Discr_TaxResDecr$ coefficient is insignificant. Together, these results suggest that discretionary increases in the deferred tax valuation allowance and the reserve for uncertain tax positions allow managers to increase their GAAP ETRs to imitate a significant IPO without losing cash tax savings. However, managers only appear to use reductions in the valuation allowance and not the reserve for uncertain tax benefits to decrease their firm's GAAP ETRs. In our Table 8 Panel D regressions, we only document one instance where the $Diff_LagETR_IPO$ coefficient is negative and significant at conventional levels (i.e., $p < 0.01$) and one instance where it is marginally significant (i.e., $p < 0.20$). This result suggests that a substantial portion of incumbent GAAP ETR herding is through tax accruals.²⁴

5. Additional Analyses and Robustness Tests

5.1. Summary of Online Appendix

Our online appendix provides detailed analyses ruling out several alternative explanations for our results and demonstrating their robustness. A potential concern is that our primary results are attributable to other nontax changes in corporate policy that lead to secondary tax effects. We document that our results are robust to controlling for the distance between IPO and incumbent firms' scaled pretax cash flows and the distance between IPO and incumbent firms' pretax return on assets. We also rule out the possibility that our results are attributable to nontax issues. One nontax issue is competition from the significant IPO that can diminish incumbent operating performance and indirectly affect the ETR. We estimate the main model using peer incumbents as those in the same MSA as the significant IPO. Although IPOs negatively affect peer firms in the same industry due to competitive effects, an IPO is less likely to directly compete with other firms in its MSA (relative to other firms in its industry). Our evidence is consistent with incumbent firms imitating the significant IPOs in their MSA.

Although we control for the distance between an incumbent firm's ETR and the industry median, it is still possible that our results reflect an incumbent firm's attempt to imitate an industry benchmark that is near the significant IPO's ETR. We form partitions that only contain incumbents and significant IPOs above the industry median ETR or only contain incumbents and significant IPOs below the industry median ETR. If the herding motivation describes the movement in ETRs, we would observe incumbent firm ETRs move toward the significant IPO's ETR while also moving away from the industry median.

Our results are consistent with a herding explanation. Another alternative explanation is that incumbent firms "learn" from the financial reports of IPO firms. If the learning explanation were descriptive, we would be more likely to document imitation when IPOs have higher quality accounting. We use three measures of accounting quality in separate regressions to test for cross-sectional variation in the herding response based on the accounting quality of the IPO firm. We do not find variation in the herding response for the accounting quality of the significant IPO firm, further supporting a herding explanation. In a final attempt to rule out alternative explanations, we acknowledge that our statistical tests rely on a sample of identified significant IPO events within an industry and an affected group of incumbent firms within the same industry. Our research design's inherent feature is that it does not allow a counterfactual response to compare against the incumbent firm response we document. We add additional support for our primary results using a control sample approach and the IPWRA estimation procedure.²⁵

5.2. Untabulated Additional Analysis

Schwab et al. (2022) find that for firms with extreme GAAP ETRs, factors unrelated to tax avoidance (e.g., operating performance) explain the most deviation in the GAAP ETRs from the statutory tax rate. Our focus is on changes in ETRs and the distance between incumbent firms' ETRs and the significant IPO's ETR rather than the distribution of ETRs. However, we winsorize GAAP ETR changes at the 10th and 90th percentiles, respectively, and re-estimate Equation (1) consistent with Schwab et al. (2022). We also use IPWRA using the winsorized GAAP ETR sample. To further mitigate the effect of potential, influential observations, we employ robust regression (Leone et al. 2019) and re-estimate Equation (1). In all three sets of analyses, our inferences remain unchanged. We also restrict our analysis to the subsample of incumbent firm-years associated with significant IPO year $t - 1$ ETRs that represent reasonable benchmarks toward which the incumbent firm would herd ($0 < \text{IPO year } t - 1 \text{ ETR} \leq 0.40$) and find our inferences unchanged. We control for prior year changes in incumbent firms' ETRs to ensure our results are not attributable to mean reversion in incumbent firms' ETRs and find our results unchanged. Our results are also robust to including actual and expected future pretax income as control variables.

6. Conclusion

Prior research provides evidence on the factors influencing the decision to go public, pricing patterns surrounding the IPO event, and the long-run performance of the IPO itself. However, an IPO firm is only one firm

in the industry in which it operates. More economically significant to aggregate industry outcomes is the effect of the IPO on the existing publicly traded firms. The few studies that examine the effect of the IPO on incumbent firms find that a significant IPO event in an industry adversely affects the industry's incumbent firms' market values, returns, and operating performance. However, no studies examine how incumbent firms' corporate policies respond to IPO events, if at all. We examine whether incumbent firms' tax policy decisions imitate significant IPOs' disclosed tax policy outcomes.

We consistently find that incumbent firms herd toward the tax outcomes of significant IPO firms. Importantly, our results are not limited to tax minimization activities but suggest that firms will increase and decrease reported tax outcomes to imitate a significant IPO, consistent with costs associated with the perception of the incumbent as a tax-avoiding firm. Attention effects underlying the herding theory of imitation and tax-specific focus on incumbent firm policies exacerbate the herding response. We also document variation in the herding response consistent with the costs of adjusting tax expense. Our results suggest that tax accrual adjustments are one means incumbent firms use to achieve immediate herding toward the significant IPO's tax policies, and imitation using cash tax planning occurs over a longer period following the significant IPO. We substantiate that our results occur because of significant IPO events and not a broader imitation of the significant IPO and incumbent firms to a common industry benchmark.

This study contributes primarily to two streams of research. First, we contribute to the small but growing literature on the effect of IPOs on the overall industry in which they operate. These studies find that an IPO adversely affects the market value and subsequent operating performance of incumbent firms in the industry (Hsu et al. 2010, Chod and Lyandres 2011). However, they do not examine whether the IPO firm and its effect on the incumbent firm market and operating performance alter incumbent firms' management decisions. We extend this research by examining whether a significant IPO changes incumbent firms' corporate tax policies, which are significant and material to individual firms.

Second, we contribute to tax literature by examining the effect of peer firms on corporate tax decisions. Prior studies suggest that firms increase tax avoidance activities in response to industry leaders' tax planning (Kubick et al. 2015). They also suggest firms adjust their ETRs in response to peer tax rate changes relating to executive turnover (Bird et al. 2018) and adjust their ETRs to foreign and state tax law changes (Armstrong et al. 2019). By contrast, we ground our expectations in the herding theory of imitation. Although prior literature finds evidence that firms imitate the tax minimizing behaviors of peer firms, they focus primarily on settings where rivalry motives, not herding, dominate or where it is difficult or impossible to disentangle between the two imitation theories. We find evidence of herding motivated imitation that represents a "leader-follower" response in industry ETRs, as opposed to rivalry-motivated imitation or a "general diffusion" of tax policy (Armstrong et al. 2019).

We also introduce to the tax literature evidence that the existence of a significant IPO event is a common catalyst for a change in incumbent firms' tax policies. We provide evidence that a significant IPO can affect incumbent firms' tax policies and become an important benchmark that stakeholders use to compare incumbent firms. Our results should also be of interest to investors because we provide evidence that a significant IPO occurring within the industry in which they invest can affect the industry's existing public firms' corporate policies, underscoring the importance of significant IPOs to capital markets. Finally, prior studies find that firms use the deferred tax valuation allowance (Frank and Rego 2006) and the reserve for uncertain tax positions (Cazier et al. 2015) to achieve earnings management objectives. We extend this literature with evidence that firms also use these accounts to imitate the tax outcomes of significant industry IPO firms in the industry.

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Appendix. Variable Definitions

Variable	Definition
Main variables	
<i>CASHETR</i>	Cash ETR, equals Compustat TXPD/(PI-SPI).
<i>GAAPETR</i>	GAAP ETR, equals Compustat TXT/(PI-SPI).
$\Delta CASHETR$	Change in <i>CASHETR</i> from year $t - 1$ to year t .
$\Delta GAAPETR$	Change in <i>GAAPETR</i> from year $t - 1$ to year t .
<i>LagETR_IPO</i>	The significant IPO firm's cash (GAAP) ETR ("ETR") in year $t - 1$.
<i>Diff_LagETR_IPO</i>	Incumbent firm's cash (GAAP) ETR in year $t - 1$ minus the significant IPO firm's cash (GAAP) ETR in year $t - 1$.
<i>IPONEWS_High</i>	An indicator: one if the amount of news on a significant IPO firm during the six months before the IPO date is higher than the median of the news on all event IPOs, zero otherwise.
<i>ImplicitETR_FC</i>	An indicator: one for incumbents whose analysts provide both a pretax forecast (measure = PRE) and an after-tax forecast (measure = NET).
<i>2002_2004</i>	An indicator; one for years 2002–2004 due to the Job Creation and Worker Assistance Act of 2002, the Jobs and Growth Tax Relief Reconciliation Act of 2003, and the American Jobs Creation Act of 2004.
<i>2009</i>	An indicator; one due to the enactment of the American Recovery and Reinvestment Tax Act of 2009.
<i>2012</i>	An indicator; one for 2012 due to the American Taxpayer Relief Act.
<i>2017</i>	An indicator; one due to the enactment of the Tax Cuts and Jobs Act of 2017.
<i>Public_Private_Public</i>	An indicator: one for incumbent firm-years associated with IPO's that were previously public within five years of the IPO year in our sample.
<i>Fin_Constraint</i>	An indicator; one if an incumbent firm-year exhibits an above median Altman (1968) Z-score. $Z\text{-score} = -1 \times \{3.3 \times [(PI + XINT)/AT] + 1.2 \times (WCAP/AT) + (SALE/AT) + 1.4 \times (RE/AT) + 0.6 \times [(CSHO \times PRCC_F)/LT]\}$
<i>Investor Attention</i>	The natural logarithm of an incumbent's Google Trends index for the month of the 10-K filing.
<i>Herd_CashETR</i>	An indicator: one if the incumbent firm is in the bottom quartile of the product of $\Delta CashETR \times Diff_LagETR_IPO$ (i.e., firms with the most negative values, indicating more movement toward the IPO's ETR), zero if in the top quartile.
<i>Herd_GAAPETR</i>	An indicator: one if the incumbent firm is in the bottom quartile of the product of $\Delta GAAPETR \times Diff_LagETR_IPO$ (i.e., firms with the most negative values, indicating more movement toward the IPO's ETR), zero if in the top quartile.
<i>%Change Analyst Following</i>	The percentage change in an incumbent firm's analyst following from year t to year $t + 1$.
$\Delta Non\text{-}Tax\ FN\ Cosine\ Similarity$	The year $t - 1$ to year t change in the cosine similarity of an incumbent's nontax footnote disclosures to the significant IPO's year $t - 1$ nontax footnote disclosures. Obtained from SeekEdgar.
$\Delta Tax\ FN\ Cosine\ Similarity$	The year $t - 1$ to year t change in the cosine similarity of an incumbent's tax footnote disclosures to the significant IPO's year $t - 1$ tax footnote disclosures. Obtained from SeekEdgar.
<i>Discr_VA</i>	Discretionary change in the valuation allowance; the reported valuation allowance change minus the expected valuation allowance change estimated by year and industry (one-digit SIC code). We follow the approach used by Frank and Rego (2006), augmented to include an additional year of future income change.
<i>Discr_VADecr</i>	An indicator: one if <i>Discr_VA</i> is negative, zero otherwise.
<i>Discr_VAIncr</i>	An indicator: one if <i>Discr_VA</i> is positive, zero otherwise.
<i>Discr_TaxRes</i>	Discretionary change in the reserve for uncertain tax positions, calculated as the reported tax reserve change minus the expected tax reserve change estimated by year and industry (one-digit SIC code). We follow the model in Gupta et al. (2016).
<i>Discr_TaxResDecr</i>	An indicator; one if <i>Discr_TaxRes</i> is negative, zero otherwise.
<i>Discr_TaxResIncr</i>	An indicator; one if <i>Discr_TaxRes</i> is positive, zero otherwise.
<i>Diff_LagCFO_IPO</i>	Incumbent firm's pretax operating cash flow in year $t - 1$ minus the significant IPO firm's pretax operating cash flow in year $t - 1$. Operating cash flow is (OANCF–XIDOC+TXPD), scaled by average total assets.
<i>Diff_LagROA_IPO</i>	Incumbent firm's pretax return on assets in year $t - 1$ minus the significant IPO firm's pretax return on assets in year $t - 1$. Pretax return on assets is equal to PI scaled by average total assets.
<i>IPO_HighDAcc</i>	An indicator: one if the signed value of performance adjusted abnormal accruals of the associated IPO in year $t - 1$ is above the median.
<i>IPO_HighAbsDAcc</i>	An indicator: one the absolute value of performance-adjusted abnormal accruals of the associated IPO in year $t - 1$ is above the median.
<i>IPO_HighFSD</i>	An indicator: one if the financial statement divergence score of the IPO in year $t - 1$ is above the median.
<i>Below_IPO</i>	An indicator: one if an incumbent firm's cash (GAAP) ETR in year $t - 1$ is lower than the significant IPO firm's cash (GAAP) ETR in year $t - 1$.
<i>Above_IPO</i>	An indicator: one if an incumbent firm's cash (GAAP) ETR in year $t - 1$ is higher than the significant IPO firm's cash (GAAP) ETR in year $t - 1$.
Control variables	
<i>Diff_LagETR_Median</i>	Incumbent firm's cash (GAAP) ETR in year $t - 1$ minus industry median cash (GAAP) ETR in year $t - 1$ (<i>ETR_Median</i>).
ΔRD	Change in research and development expense (RD). Research and development expense equals Compustat XRD scaled by average total assets, where XRD is replaced with zero if missing.

Appendix. Variable Definitions (Continued)

Variable	Definition
ΔAD	Change in advertising expense (<i>AD</i>). Advertising expense equals Compustat XAD scaled by average total assets, where XAD is replaced with zero if missing.
ΔSGA	Change in selling, general, and administrative expense (<i>SGA</i>). Selling, general, and administrative expense is Compustat XSGA scaled by average total assets, where XSGA is replaced with zero if missing.
$\Delta CAPEXP$	Change in capital expenditures (<i>CAPEXP</i>). Capital expenditures equal Compustat CAPX scaled by average total assets, where CAPX is replaced with zero if missing.
$\Delta CASH$	Change in cash and cash equivalents (<i>CASH</i>). Cash and cash equivalents is Compustat CHE scaled by average total assets.
ΔFI	Change in foreign income (<i>FI</i>). Foreign income equals Compustat PIFO scaled by average total assets, where PIFO is replaced with zero if missing.
$\Delta EQINC$	Change in equity method earnings (<i>EQINC</i>). Equity method earnings equal Compustat ESUB scaled by average total assets, where ESUB is replaced with zero if missing.
ΔNOL	Change in <i>NOL</i> where <i>NOL</i> is an indicator variable that equals one if Compustat TLCF is positive, zero otherwise.
$\Delta INTAN$	Change in intangible assets (<i>INTAN</i>). Intangible assets are equal to Compustat INTAN scaled by average total assets.
ΔPPE	Change in property, plant, and equipment (<i>PPE</i>). Property, plant, and equipment is Compustat PPENT scaled by average total assets, where PPENT is replaced with zero if missing.
ΔFCF	Change in free cash flow (<i>FCF</i>). Free cash flow equals Compustat (OANCF – CAPX) scaled by average total assets.
$\Delta SIZE$	Change in firm size (<i>SIZE</i>). Firm size is the natural logarithm of the beginning of year Compustat AT.
ΔROA	Change in pretax return on assets (<i>ROA</i>). Pretax return on assets equals Compustat PI scaled by average total assets.
ΔLEV	Change in leverage (<i>LEV</i>). Leverage equals Compustat DLTT scaled by average total assets.
<i>Attention_PrevYear</i>	The natural logarithm of the 12-month average Google Trends index in the year before the 10-K filing.
<i>BHAR_10kmonth</i>	The absolute value of the buy and hold return, adjusted for value-weighted market portfolio return, in the month of the 10-K filing.
<i>ShareTurnover</i>	Share turnover in the month of the 10-K filing.
<i>AnalystFollowing</i>	Natural logarithm of the firm's analyst following for the current year.
<i>MVE</i>	The market value of equity.
<i>MTB</i>	Market to book ratio.
ΔMTB	Change in <i>MTB</i> from year $t - 1$ to year t .
$\Delta SecurityIssuance$	An indicator variable for a large debt or equity security issuance in the current or previous year.
$\Delta Return$	Change in annual buy and hold return.
$\Delta ShareTurnover$	Change in <i>ShareTurnover</i> .
$\Delta Return_Volatility$	Change in return volatility.
IPO variables	
<i>GROSSPROCEEDS</i>	IPO gross proceeds, calculated as (IPO price $\times$ Shares offered).
<i>INITIALRETURN</i>	Initial return, calculated as (Closing price on the first day of trading-IPO price)/IPO price, in percentage.
<i>UNDERWRITER_RANK</i>	Lead underwriter's ranking of the significant IPO firm (Carter et al. 1998, Loughran and Ritter 2004), where nine is the highest and one is the lowest rank.
<i>BIGN</i>	An indicator: one if a Big N auditor audits the significant IPO firm in the year before the IPO.
<i>TOTALASSETS_PreIPO</i>	Total assets of the significant IPO firm in the year before the IPO.
<i>ROA_PreIPO</i>	Compustat PI scaled by total assets of the significant IPO firm in the year before the IPO.

Endnotes

¹ For example, Facebook Inc., one of the IPOs that we examine, raised \$16 billion and was mentioned in nearly 2,500 articles in the six months leading up to its IPO.

² See Ritter and Welch (2002) for a review of the early literature. Recent examples of studies examining IPO firms include Willenborg et al. (2015), Barth et al. (2017), Jenkinson et al. (2018), Dambra et al. (2019), Bushee et al. (2020), and Liu and Wu (2021).

³ Like Hsu et al. (2010), we focus on a subset of IPOs that raise the largest amounts of capital and, therefore, generate the largest amounts of attention ("significant IPOs"). We focus on significant IPOs because they are the most likely to elicit a herding response in incumbent firms.

⁴ Firms must balance the maximization of after-tax cash flows with the costs associated with doing so through tax avoidance. Practitioner

guides warn clients of the potential risks arising from establishing a reputation as a tax-avoiding company. In addition, the media and activist groups also target firms that appear to engage in tax avoidance. We discuss the costs of tax planning in detail in Section 2.

⁵ We choose the ETR because it the most visible summary measure of tax policy choices, and it is unlikely that peer firms would imitate other academic proxies for tax policy choices (e.g., book-tax differences or the Henry and Sansing (2018) measure of cash tax preferences). We acknowledge that ETRs have known sources of measurement error and we address them in robustness testing.

⁶ Chod and Lyandres (2011) also examine the effect of a firm's IPO on the market value of its industry rivals. However, the focus of their predictions and empirical analyses are on the going public decision of a private firm and the market strategies of the IPO firm following the IPO.

⁷ We address the effects of operating performance on our results using several distinct methods in the online appendix and summarized in Section 5 and consistently document a tax-specific herding response in incumbent firms.

⁸ Pollock et al. (2008) explain that the IPO setting is most characterized by herding motives, particularly those driven by availability cascades, because of its inherent uncertainty, where uncertainty is “environmental uncertainty” in that managers have weak prior probabilities about the likely success of alternative paths and are more open to external sources of information. In contrast, rivalry-based motives are more likely to dominate when firms’ knowledge of rivals’ products and processes is relatively high. We rely primarily on the herding theory of imitative behavior because it is most appropriate for our IPO setting. In untabulated analysis, we find that our results are not explained by rivalry motives. See both Lieberman and Asaba (2006) and Pollock et al. (2008) for a detailed discussion of the theories of imitation and discussion of environmental uncertainty.

⁹ See https://assets.ey.com/content/dam/ey-sites/ey-com/en_gl/topics/growth/ey-guide-to-going-public-strategic-considerations-before-during-and-post-ipo.pdf (accessed December 1, 2018).

¹⁰ See <https://www.pwc.com/gx/en/tax/publications/assets/pwc-tax-strategy-and-corporate-reputation-a-tax-issue-a-business-issue.pdf> (accessed December 1, 2018), or https://www.eytax.jp/tax-library/other-report/pdf/Tax-Risk-and-Controversy-Survey-2014_E.pdf (accessed December 1, 2018).

¹¹ From this list, we drop any IPOs of publicly traded partnerships that elect to be taxed as a pass-through entity because their tax rates and incentives will not be comparable to our sample of incumbent firms.

¹² Edwards et al. (2019) examine the pricing and performance of 50 “supercharged” IPOs. To be a supercharged IPO, the entity must be a passthrough entity prior to the IPO and would therefore have no entity level tax pre-IPO ETR. Such IPO firms are excluded from our sample.

¹³ Most IPOs generate tax losses prior to the IPO year (Allen 2019). In untabulated analysis, we find that before exclusion, 27% of the significant IPOs have negative pretax book income before special items, whereas the corresponding ratio for non-significant IPOs during the same sample period is 45%. Thus, our focus on significant IPOs results in the exclusion of relatively fewer IPO firms with negative book income.

¹⁴ Our IPO events begin in 1993 and our matched incumbents enter the sample in 1994. The list of significant IPOs in our sample, their IPO year, and their two-digit SIC code are provided in the online appendix, Table EC.1.

¹⁵ In untabulated analysis, we confirm the robustness of our results to matching significant IPOs and incumbent firms using three-digit SIC codes. In Section 5, we also discuss the results of matching significant IPOs with incumbent firms in the same MSA.

¹⁶ We acknowledge the possibility that firms may be unable to imitate the new significant IPO and instead changes in their ETRs represent imitation of an alternative industry benchmark (e.g., the industry median) or natural reversion to the industry median over time. By controlling for the industry median ETR, β_1 can be interpreted as changes in an incumbent firm’s ETR that cannot be explained by reversion to the industry median. We perform several additional tests to further rule out the possibility that our results only represent a reversion to industry medians in Section 5.

¹⁷ In untabulated additional analyses, we also control for expected performance by including analysts forecasts of one year ahead income and, separately, $\Delta ROA_{i,t+1}$. We continue to find support for our hypothesis.

¹⁸ We also test whether incumbents’ herding behavior is incremental to the mimicking of industry leaders by including the distance between the incumbent ETR and the industry leader’s ETR ($Diff_Lag_ETR_LEADER$) following the Kubick et al. (2015). In untabulated results, we find a negative and significant coefficient on $Diff_Lag_ETR_IPO$ ($\Delta GAAPETR$: -0.033 , $t = -3.79$; $\Delta CASHETR$: -0.018 , $t = -2.17$). The coefficient on $Diff_Lag_ETR_LEADER$ is highly significant (negative).

¹⁹ We necessarily exclude $Diff_Lag_ETR_Median$ (the difference between the incumbent firm’s ETR and the industry median) in this estimation.

²⁰ Standard deviation of $Diff_Lag_ETR_IPO$ for $Low_ETR = 1$ incumbents of 0.1633 times the sum of the estimated main effect and interaction ($-0.042 + -0.271$) ≈ -0.051 . Standard deviation of $Diff_Lag_ETR_IPO$ for $Low_ETR = 0$ incumbents of 0.1357 times the estimated main effect (-0.042) ≈ -0.006 .

²¹ We do not imply that the adjustment of tax accruals is costless because firms face constraints with respect to tax accrual management, including adherence to GAAP and auditor, investor, and regulatory oversight. To understand how incumbents imitate tax policy, we examined the 10-K disclosures of 169 incumbent firm years with a year $t - 1$ ETR in the same quintile as the significant IPO year $t - 1$ ETR ($Same_Quint = 1$) and exhibit a herding response of five or fewer percentage points. We present in Table EC.2 of the online appendix examples of incumbent firms ETR rate reconciliations and associated text explanations, where provided, from the sample. We chose two examples of increases in incumbent ETRs to herd toward the IPO and two examples of decreases in incumbent ETRs to herd toward the IPO. The most common explanation for GAAP ETR changes is tax accrual changes, consistent with our expectations.

²² In untabulated analyses, we focus on the subset of incumbent firm-years that end within six months of the IPO date. The ability to alter pretax, or even significant cash tax policies, is even further constrained. We continue to find a significant GAAP ETR herding response and no cash ETR herding response.

²³ Frank and Rego (2006) control for changes in pretax income in $t - 1$, t , and $t + 1$. We include these and $t + 2$. Our results are qualitatively and quantitatively similar if we control for two additional years of changes (i.e., if we also add $t + 3$).

²⁴ In untabulated analysis, we estimate Equation (1) on the subset of firms that record a zero balance for net operating losses. We continue to find evidence suggesting firms manage their GAAP ETRs upward and downward to move closer to the focal IPO firm, even when their ability to do so through valuation allowance adjustments is substantially limited. Furthermore, we find that the magnitude of the $Diff_Lag_ETR_IPO$ coefficient is not statistically different between the partitions suggesting the magnitude of the herding response is similar for upward and downward herding in this subsample of firms.

²⁵ Please refer to the Online Appendix, Austin (2011), and Austin and Stuart (2015) for more details surrounding the IPWRA process.

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